

Power Law Alignment

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1 Introduction

Graph alignment, also known as graph matching, seeks to uncover a hidden bijection between the vertex sets of a pair of similar graphs. In this thesis, we consider the following formulation of the alignment problem for correlated random graphs.

Problem 1.1. *Let G be a random graph with vertex set $[n]$, and let G_1 and G_2 denote two independent edge-subsamplings of G ; that is to say, retain each edge $(u, v) \in E(G)$ in the set $E(G_i)$ with probability s . Sample a permutation $\pi^* \in S_n$ uniformly at random and permute the vertex labels of G_2 according to π^* to obtain G'_2 . Given G_1 and G'_2 , recover the permutation π^* .*

We will be interested in the performance of estimators $\hat{\pi}$ of π^* as the number of vertices n tends to ∞ , so the parameter s is allowed to depend on n . The graph G is also referred to as the parent graph from which the two children G_1 and G_2 are sampled, the subsampling parameter s is also known as the correlation parameter, and the permutation π^* is also called the planted matching. Without loss of generality, we often take π^* to be the identity permutation, so that G_2 and G'_2 may be identified. In this case one should imagine that the vertex labels of G_1 and G_2 have been erased. Note that in the literature there are a number of alternative versions of the graph alignment problem; for instance, Bayati et al. (2013) incorporates additional weights to encode similarity scores between pairs of vertices in G_1 and G_2 . However, the formulation given in Problem 1.1 is the simplest setting in which one can discuss theoretical results on alignment, which is the focus of this thesis.

Since the graphs G_1 and G'_2 are random, there are multiple notions of recovery in the alignment problem. We follow the conventions of Ganassali et al. (2022). Consider a proposed (partial) matching $\hat{\pi} : \mathcal{C} \rightarrow [n]$, where $\mathcal{C}$ is a subset of $[n]$. The overlap of $\hat{\pi}$ with π^* is the fraction of vertices it correctly matches:

$$\text{ov}(\pi^*, \hat{\pi}) = \frac{1}{n} \sum_{u \in \mathcal{C}} \mathbf{1}_{\hat{\pi}(u) = \pi(u)} \quad (1)$$

The error fraction of $\hat{\pi}$ is the fraction of vertices it incorrectly matches:

$$\text{err}(\pi^*, \hat{\pi}) = \frac{1}{n} \sum_{u \in \mathcal{C}} \mathbf{1}_{\hat{\pi}(u) \neq \pi(u)} \quad (2)$$

An estimator achieves exact recovery if $\hat{\pi}(u) = \pi^*(u)$ for every $u \in [n]$ whp. An estimator achieves partial recovery if there exists a constant $\eta > 0$ such that $\text{ov}(\pi^*, \hat{\pi}) > \eta$ whp.

Finally, an estimator achieves one-sided partial recovery if it achieves partial recovery and if $\text{err}(\pi^*, \hat{\pi}) = o(1)$ whp. We will primarily be concerned with sparse random graphs G – those with constant average degree. Such graphs are common in modeling real world networks, from user preference databases Narayanan and Shmatikov (2008), to social networks Narayanan and Shmatikov (2009), to protein-interaction networks Jensen et al. (2009). Results from information theory show that in some models, exact recovery is impossible in this regime Cullina and Kiyavash (2016), so we restrict attention to estimators $\hat{\pi}$ that achieve partial or one-sided partial recovery of π^* .

The graph alignment problem is motivated by a variety of applications in the natural and social sciences. As we shall see in the examples below, graph alignment generally allows one to transfer knowledge between two related contexts.

Protein-protein interaction networks: Proteins are essential components of living cells. Through proteomics, the study of the collection of all proteins within a cell, biologists can derive greater insights than through genomics alone Graves and Haystead (2002). Of particular relevance to graph alignment is the study of protein-protein interaction (PPI) networks, where vertices are proteins and edges are biological interactions between them. The structure of such PPI networks has been investigated by various authors, and in particular Yook et al. (2004) revealed scale-free properties of a yeast PPI network. An early example of alignment of PPI networks is Singh et al. (2007), in which the authors introduce and apply the IsoRank algorithm (a relaxation of a weighted quadratic program) to match a yeast PPI network to a fruit fly PPI network, deriving evolutionary insights from the results. In Bayati et al. (2013), the authors develop the message passing algorithm NetAlignMP and use it to again align yeast and fruit fly PPI networks, as well as human and mouse PPI networks. Further progress was made by Kazemi et al. (2016), who introduced the seeded percolation-based PROPER algorithm based on the theoretical results in Kazemi et al. (2015).

Deanonimization: Another well-known application of graph alignment is the deanonymization of datasets. This line of work was initiated by the now-famous paper Narayanan and Shmatikov (2008). The authors began by formulating a model of sparse databases of user preferences, and then applied their analysis to the Netflix Prize dataset, which had been anonymized and released to the public for the sake of a machine learning competition. Remarkably, Narayanan and Shmatikov (2008) found that for 99% of users, an adversary could uncover their identity with merely eight of their movie ratings. With the help of 50 public user records on the Internet Movie Database, the authors were able to identify a number of Netflix user profiles with high confidence. In the follow-up work Narayanan and Shmatikov (2009), the authors extended their ideas to social networks, demonstrating that a substantial fraction of an anonymized Twitter dataset could be determined through seeded alignment with the Flickr user network. Social networks have grown exponentially since these findings, and in the wake of scrutiny over the data-sharing practices of social media companies, graph alignment remains highly relevant Ji et al. (2016).

Many but not all of the algorithms referenced above use certain heuristics to perform alignment. In this thesis we focus on graph alignment algorithms with theoretical guarantees. As a result, we will restrict our attention to certain stylized random graph models.

Power law random graphs are inhomogeneous random graphs that capture certain features of real world networks Barabási and Bonabeau (2003). We begin by discussing the relevance

of power law models to real world networks. We then survey existing results on the alignment of power law random graph models. Finally, we define our own stratified power law random graph model and present a seedless alignment algorithm that correctly matches a constant fraction of the vertices whp. The first stage of our algorithm involves finding seeds, and the remaining stages involve propagating the matching. We note that our algorithm for finding seeds is very similar to the degree profile-based procedures introduced in Ding et al. (2021) for the Erdős–Rényi model and generalized in Ding et al. (2023) to an inhomogeneous model. However whereas Ding et al. (2023) requires the correlation s to tend to 1, we work in the setting of constant correlation $s \in (0, 1]$. Our algorithm for propagating the matching is quite standard in the percolation graph matching literature, and in particular is very similar to arguments in Korula and Lattanzi (2013).

Throughout this thesis, we employ some common notation: o , O , ω , and Ω are standard Landau notation; $f \sim g$ if $f/g = 1 + o(1)$; “wp” stands for “with probability”; a sequence of events $(A_n)_{n \in \mathbb{N}}$ holds with high probability (“whp”) if the probability of A_n tends to 1 as n goes to ∞ ; “iid” stands for independent and identically distributed; $\stackrel{d}{=}$ and $\stackrel{d}{\rightarrow}$ denote equality and convergence in distribution, respectively; $\stackrel{p}{\rightarrow}$ denotes convergence in probability; $\stackrel{as}{\rightarrow}$ denotes almost sure convergence; $d_{TV}(\mu, \nu)$ denotes the total variation distance between probability measures μ and ν ; the moment generating function (“mgf”) of a random variable X is the function $\theta \mapsto \mathbb{E}[e^{\theta X}]$; and Bern, Binom, Geom, Pois, Exp, Gamma, and Unif denote the Bernoulli, Binomial, Geometric, Poisson, Exponential, Gamma, and Uniform distributions, respectively.

2 Real world networks

The Erdős–Rényi model of Erdős et al. (1960) was introduced in 1960, and is perhaps the oldest and best-understood of all random graph models. However, as was quickly realized in the early days of network science, the Erdős–Rényi model is inadequate in describing many aspects of real world networks. For instance, the empirical degree distribution of real world networks is often heavy-tailed, whereas the degrees in the Erdős–Rényi model are Poissonian with exponential tails. One of the most famous papers in the field of network science is Barabási and Albert (1999), in which the authors propose the Barabási–Albert model to generate random graphs with a power law degree distribution; that is to say, the degree D of a uniformly random vertex obeys $\mathbb{P}(D = k) \propto k^{-\beta}$ for some $\beta > 0$. (Typically, β is taken to lie in $(2, 3)$ Barabási and Bonabeau (2003).) The Barabási–Albert model is a dynamic model such that when a new vertex is added to the graph, it is more likely to connect to existing vertices with higher degrees. This preferential attachment mechanism leads to the power law structure, as well as the high-degree hubs characteristic of such scale-free graphs Barabási and Bonabeau (2003). In their article, Barabási and Albert (1999) suggested that preferential attachment is a unifying principle for the ubiquity of power laws, which have also been observed in the World Wide Web Barabási et al. (2000), PPI networks Yook et al. (2004), citation networks Redner (1998), and semantic networks like WordNet Steyvers and Tenenbaum (2005). Later, other models were shown to give rise to power law degree distributions, such as the Chung-Lu model, which assigns a positive weight w_v to each vertex v in the graph and independently includes edge (u, v) in $E(G)$

with probability $1 \wedge \frac{w_u w_v}{\sum_{k \in [n]} w_k}$ van der Hofstad (2017). For a particular choice of weights, the power law degree distribution can be reproduced, as demonstrated in Chung and Lu (2002). While the importance of heavy-tailed distributions such as the power law has been widely acknowledged by the community, there has been skepticism regarding the universality of any particular distribution. Of particular note is the paper Clauset et al. (2009), which points out ambiguities and inaccuracies in established methods of fitting power laws to degree distributions. For one, it is obviously necessary to cut-off the power law at certain minimum and maximum values, but it is not at all clear how to select this range. In Clauset et al. (2009), the authors construct a goodness-of-fit test and reject the plausibility of a power law in 7 of 24 networks, including the network of links on the World Wide Web. Although the notion of universality continues to be hotly debated, in this thesis we consider a power law model merely as a stepping stone to a deeper theoretical understanding of alignment in inhomogeneous networks.

3 Literature review: power law alignment

We survey a few algorithms for alignment of correlated random graphs with power law degree distributions.

Most algorithms in this setting require a small set of “seeds”, a few revealed pairs $(v, v) \in V(G_1) \times V(G_2)$ of correctly matched vertices. To propagate the matching, such an algorithm decides whether to match a potential pair $(v, u) \in V(G_1) \times V(G_2)$ based on the proximity of (v, u) to pairs that have already been matched. A heuristic seeded alignment algorithm was introduced in the early work Narayanan and Shmatikov (2008) to match the Twitter and Flickr user networks. In Yartseva and Grossglauser (2013), the authors use percolation theory to analyze the following PercolateMatched algorithm in the Erdős–Rényi setting. Initially, a set of seed pairs is revealed and added to the matching. The score of a vertex pair $(v, u) \in V(G_1) \times V(G_2)$ is equal to the number of neighboring previously-matched pairs $(v', u') \in V(G_1) \times V(G_2)$; that is to say, the number of previously-matched pairs (v', u') such that $(v, v') \in E(G_1)$ and $(u, u') \in E(G_2)$. Such pairs (v', u') are also referred to as “1-hop witnesses”. At each stage, PercolateMatched adds a pair (v, u) to the matching if its score exceeds a given threshold r . In Kazemi et al. (2015), the authors refine these ideas and introduce NoisySeeds and ExpandWhileStuck, a pair of algorithms that allow for robust graph matching with fewer seeds, with theoretical guarantees in the Erdős–Rényi case. Unlike PercolateMatched, ExpandWhileStuck does not immediately include the potentially incorrect seed pairs in the final matching, replacing these seeds with a larger set of candidate pairs. The score of a vertex pair is equal to the number of neighboring candidate pairs, and a vertex pair is added to the matching when this score is at least r .

Seeded alignment techniques are especially attractive in the power law setting, due to the high-degree hubs present in scale-free networks Barabási and Bonabeau (2003).

In Korula and Lattanzi (2013), the authors propose and analyze the UserMatching algorithm, an efficient seeded percolation-based alignment algorithm for the scale-free Barabási–Albert model. At a given stage j , a vertex pair $(v, u) \in V(G_1) \times V(G_2)$ is added to the matching iff the degrees of v and u are both at least 2^j and if u is the vertex with the most 1-hop witnesses in common with v . The degree threshold is gradually decreased as the

matching progresses, so that lower degree vertices are matched later in the process. The UserMatching algorithm requires $\Omega\left(\frac{n}{\log n}\right)$ seeds.

In Bringmann et al. (2014), the authors present a seeded alignment algorithm for the Chung-Lu power law model first proposed in Chung and Lu (2002) with power law parameter $\beta \in (2, 3)$. The algorithm propagates the matching in the following manner. For each unmatched vertex pair (v, u) , the algorithm compares the neighborhoods of v and u among the already-matched vertices using a multiplicative statistic, and adds (v, u) to the matched pairs if this statistic exceeds a certain quantity. For any $\varepsilon > 0$, if provided with the n^ε highest degree vertices in the graph, the algorithm can match a constant fraction of the vertices whp in quasilinear time.

Chiasserini et al. (2016) presents another seeded alignment algorithm in the Chung-Lu power law setting with $\beta \in (2, 3)$. The degree-driven graph matching (DDM) algorithm first partitions the vertices into slices such that in each slice, the vertices have similar degrees. It begins by matching a high slice using percolation graph matching. It then inductively descends through the graph, matching vertex pairs in lower slices based on one-hop witnesses among already-matched pairs. For any $\varepsilon > 0$, if seeds are distributed uniformly among all vertices, then the DDM algorithm requires $n^{\frac{1}{2}+\varepsilon}$ seeds to match a constant fraction of the vertices. If on the other hand one can select the seeds, then only n^ε seeds are necessary.

In Yu et al. (2021), the authors obtain dramatic improvements in the Chung-Lu power law setting by going beyond 1-hop witnesses. By considering D -hop witnesses of a given vertex pair (v, u) – previously-matched pairs (v', u') such that v and v' are D -hop neighbors in G_1 , and such that u and u' are D -hop neighbors in G_2 – the authors prove that their Power Law D -hop (PLD) algorithm can correctly match a constant fraction of vertices when provided with only $\Omega((\log n)^{4-\beta})$ uniformly chosen seeds, assuming that $\beta \in (2, 3)$ and $D > \frac{4-\beta}{3-\beta}$. This also requires a careful slicing of each of the vertex sets $V(G_1)$ and $V(G_2)$.

In a separate line of work, Ding et al. (2023) propose a seedless alignment algorithm for an inhomogeneous generalization of the Erdős–Rényi model when the edge correlation is at least $1 - O\left(\frac{1}{(\log n)^2}\right)$. The signature of a vertex is a function defined on $[0, 1)$ obtained by computing the degrees of each of its neighbors, constructing a continuous family of bins, and counting the number of degrees whose square roots land into each bin. The similarity of two vertices is measured using the L^1 distance between these signature functions. This is a vast generalization of the previous work Ding et al. (2021), which studies the Erdős–Rényi setting with edge correlation at least $1 - O\left(\frac{1}{(\log n)^2}\right)$. In Ding et al. (2021), the signature of a vertex is obtained by computing the outdegrees of each of its neighbors, constructing a discrete set of bins, and counting the proportion of normalized outdegrees that land in each bin. The similarity of two vertices is measured by the ℓ^1 distance between their signature vectors.

In what follows, we present a polynomial-time seedless alignment algorithm for a particular power law random graph model. Our model falls under the general Chung–Lu framework, but is *not* the original Chung–Lu power law model introduced in Chung and Lu (2002). The first stage of our algorithm involves finding n^α seeds among high-degree vertices by comparing the sizes of their neighborhoods among lower levels of the graph and computing a mean absolute deviation statistic. Our statistic is very similar to the degree profile-based statistic in Ding et al. (2021), but whereas Ding et al. (2023) requires the correlation to tend to 1, our analysis allows us to perform matching even at constant correlation. The second stage

involves propagating the matching by counting 1-hop witnesses in previously-matched layers, and our procedure turns out to be almost identical to the analysis in Korula and Lattanzi (2013). Although the propagation techniques are very similar, we include the analysis because our model is defined in a different manner.

4 Seedless alignment algorithm

In this section, we define a random graph model with a power law degree distribution, and we propose a seedless alignment algorithm that can provably recover a constant fraction of the vertices whp in polynomial time. Our random graph model is an instance of the general Chung–Lu model as defined in Chapter 6 of van der Hofstad (2017). The Chung–Lu model generalizes the Erdős–Rényi model by allowing edge probabilities to vary. However, our graph is *not* the original Chung–Lu power law model introduced in Chung and Lu (2002).

Let $\beta \in (2, 3)$ denote the power law parameter. Our random graph G has a stratified structure. Roughly speaking, the vertices in layer k have expected degree 2^k , and layer k contains on the order of $2^{-(\beta-1)k}n$ vertices. Thus the number of vertices of degree at least x is proportional to $x^{-(\beta-1)}$.

First, we specify the structure of the vertex set. The vertex set $V(G)$ consists of $k_{\max} + 1$ layers numbered $\{0, 1, \dots, k_{\max}\}$, where the top layer is defined to be $k_{\max} = \lfloor \frac{1}{\beta-1} \log_2 n \rfloor$. Let L_k denote the set of vertices in layer k . Let N_k denote the size of L_k . Let $C_1 = \left(\sum_{k=0}^{k_{\max}} 2^{-(\beta-1)k}\right)^{-1}$. It is easy to see that $C_1 = (1 + o(1))(1 - 2^{-(\beta-1)})$. For each $k \geq 1$, we set N_k to be $N_k = \lceil C_1 2^{-(\beta-1)k} n \rceil$. For $k = 0$, we define $N_0 = n - \sum_{k=1}^{k_{\max}} N_k$. We refer to k as the “height” of the layer L_k .

The above construction ensures that the number of vertices in each layer is an integer, and that the total number of vertices equals n . Furthermore, so long as $k_{\max} - k = \omega(1)$, we have the asymptotics $N_k = (1 + o(1))C_1 2^{-(\beta-1)k}n$.

Next, we specify the edge probabilities. Assign the weight $w_v = 2^k$ to each vertex v in layer L_k . Let w_{tot} denote the sum of the weights assigned to all vertices: $w_{\text{tot}} = \sum_{k=0}^{k_{\max}} 2^k N_k$. It is easy to see that $w_{\text{tot}} = C_2^{-1}n$, where $C_2 = (1 + o(1))(1 - 2^{-(\beta-2)})C_1^{-1}$. Given vertices $u \in L_j$ and $v \in L_\ell$, we include the edge (u, v) in the edge set $E(G)$ independently of all other edges with probability $p_{\ell j}$, where

$$p_{\ell j} = 1 \wedge \frac{w_u w_v}{w_{\text{tot}}} = 1 \wedge \frac{2^\ell 2^j}{w_{\text{tot}}} \quad (3)$$

Note that we allow self-loops. When $\frac{2^\ell 2^j}{w_{\text{tot}}} \leq 1$, we simply have

$$p_{\ell j} = \frac{2^\ell 2^j}{w_{\text{tot}}} = C_2 \frac{2^\ell 2^j}{n} \quad (4)$$

As a result, when $v \in L_\ell$ is such that $\frac{2^\ell 2^{k_{\max}}}{w_{\text{tot}}} \leq 1$, the expected degree of v is exactly 2^ℓ . It is easy to see that a layer ℓ satisfies the condition $\frac{2^\ell 2^{k_{\max}}}{w_{\text{tot}}} \leq 1$ if we have the following condition ($\dagger$): $\ell \leq (\beta - 2)k_{\max} - \omega(1)$. In our algorithm, we only consider pairs of layers (L_j, L_ℓ) that satisfy condition ($\dagger$). This simplifies the analysis.

We write $G \sim PL(n, \beta)$ to denote that G is sampled from the power law model defined above. We now state the alignment problem in our setting.

Problem 4.1. *Fix a constant $s \in (0, 1]$. Let $G \sim PL(n, p)$ be sampled from the power law random graph model defined above, and let G_1 and G_2 denote two independent edge-subsamplings of G , where each edge is retained with probability s . Sample a permutation $\pi^* \in S_n$ uniformly at random and permute the vertex labels of G_2 according to π^* to obtain G'_2 . Let $L_k^{(1)}$ denote the set of vertices in L_k when regarded as a subset of $V(G_1)$, and let $L_k^{(2)}$ be defined similarly. Given G_1 and G'_2 , and given the sets $L_k^{(1)}$ and $L_k^{(2)}$ for all k , recover the permutation π^* .*

We write $(G_1, G'_2) \sim PLC(n, \beta, s)$ to denote this correlated power law model. For convenience, in the analysis we take the permutation π^* to be the identity. Note that we provide extra information in the formulation above – we will elaborate on this assumption below.

In our analysis, we show that whp, we can correctly match a constant fraction of the vertices for any correlation $s \in (0, 1]$. An illustration of the model and the stages of our algorithm is presented in Figure 1.

It is important to note that in our formulation, we reveal which vertices belong to $L_k^{(1)}$ and $L_k^{(2)}$ for each k . Although this may seem like a major simplification, note that since the degree of each vertex concentrates around its mean, and since degrees in different layers differ by an order of magnitude, it should be the case that all but the lowest layers can be distinguished whp. However, we make the assumption that the sets $L_k^{(1)}$ and $L_k^{(2)}$ are known to simplify our analysis.

We now clarify some of the terminology that we will use. We use the phrase “layer k ” and “layer L_k ” interchangeably. When we say that “we (partially) match the layer L_k ”, we mean that we match a pair of subsets $S_k^{(1)} \subseteq L_k^{(1)}$ and $S_k^{(2)} \subseteq L_k^{(2)}$. By a “(partially) matched layer” L_k , we mean that a subset of vertices $S_k^{(1)} \subseteq L_k^{(1)}$ has already been correctly matched to subset of vertices $S_k^{(2)} \subseteq L_k^{(2)}$. Since π^* is taken to be the identity permutation, the vertex sets $V(G_1)$ and $V(G_2)$ are identical, so that a correctly matched pair is of the form $(v, v) \in V(G_1) \times V(G_2)$. Whenever we write (v, u) , we treat it as an element of $V(G_1) \times V(G_2)$, so that the first entry v is a vertex in G_1 and the second entry u is a vertex in G_2 . Given a partially matched layer L_k , where $S_k^{(1)} \subseteq L_k^{(1)}$ has previously been correctly matched to $S_k^{(2)} \subseteq L_k^{(2)}$, a “common neighbor” of vertices $v \in L_k^{(1)}$ and $u \in L_k^{(2)}$ among the correctly matched vertices of L_k is simply a pair (w, w) in $S_k^{(1)} \times S_k^{(2)}$ such that $(v, w) \in E(G_1)$ and $(u, w) \in E(G_2)$.

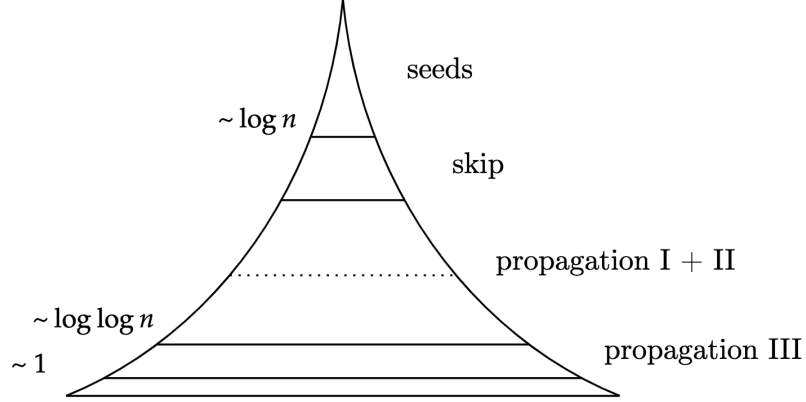


Figure 1: Schematic of our power law graph, labeled by the different algorithms we use to grow the matching. We first match seeds near the top, then propagate to $\log \log n$ height, and finally propagate to constant height.

4.1 Finding seeds

We begin by matching a few vertices with high degrees. These seeds will be used to propagate the matching.

Fix n , the total number of vertices in the graph. Given a vertex $v \in V(G)$ in the parent graph, let $X_{\ell,n}(v)$ denote the number of neighbors v has in layer L_ℓ . Given $v \in V(G_1)$, let $X_{\ell,n}^{(1)}(v)$ denote the number of neighbors v has in layer $L_\ell^{(1)}$. Given $v \in V(G_2)$, let $X_{\ell,n}^{(2)}(v)$ denote the number of neighbors v has in layer $L_\ell^{(2)}$. By the definition of our model, $X_{\ell,n}(v) \sim \text{Binom}(N_\ell, p_{\ell j})$. Since distinct edges are independent, the random variables $X_{\ell,n}(v)$ and $X_{\ell',n}(v')$ are independent so long as $\ell \neq \ell'$ or $v \neq v'$. Finally, conditional on $X_{\ell,n}(v)$, $X_{\ell,n}^{(1)}(v)$ and $X_{\ell,n}^{(2)}(v)$ are iid $\text{Binom}(X_{\ell,n}(v), s)$.

We now formulate our algorithm. Suppose that we want to match a vertex $v \in L_j^{(1)}$. If $u \in L_j^{(2)}$ is a vertex distinct from v , then $X_{\ell,n}^{(1)}(v) - X_{\ell,n}^{(2)}(v)$ has a lower variance than $X_{\ell,n}^{(1)}(v) - X_{\ell,n}^{(2)}(u)$. In other words, the number of neighbors that $v \in L_j^{(1)}$ has in $L_\ell^{(1)}$ is typically closer to the number of neighbors that $v \in L_j^{(2)}$ has in $L_\ell^{(2)}$, when compared to any mismatch $u \in L_j^{(2)}$.

Given a layer ℓ and a pair of vertices $(v, u) \in V(G_1) \times V(G_2)$, define the normalized degree-difference

$$Z_\ell^{(n)}(v, u) = \frac{1}{\sqrt{N_\ell p_{\ell j}}} (X_\ell^{(1)}(v) - X_\ell^{(2)}(u)) \quad (5)$$

Further, given a set $\mathcal{I}_n$ of layers, define the statistic

$$T^{(n)}(v, u) = \frac{1}{|\mathcal{I}_n|} \sum_{\ell \in \mathcal{I}} |Z_\ell^{(n)}(v, u)| \quad (6)$$

by the mean absolute deviation of the degree-differences.

(In our definition of $Z_\ell^{(n)}(v, u)$, we abuse notation by dropping reference to n on the right-hand side. For notational convenience, we will continue to do so below. However we will always include reference to n in $\mathcal{I}_n$, $Z_\ell^{(n)}(v, u)$, and $T^{(n)}(v, u)$.)

Our algorithm may be stated as follows. Fix a set of layers $\mathcal{J}_n$ near the top of the graph, to be specified below. Fix a set of layers $\mathcal{I}_n$, to be specified below. Suppose that we want to match a vertex $v \in L_j^{(1)}$, where $j \in \mathcal{J}_n$. For each vertex $u \in L_j^{(2)}$, compute the quantity $T^{(n)}(v, u)$ using the layers $\mathcal{I}_n$. Let $v^* \in L_j^{(2)}$ denote the minimizer $\arg \min_{u \in L_j^{(2)}} T^{(n)}(v, u)$ of this statistic. We match v^* to v . We repeat this procedure for all $v \in L_j^{(1)}$ to obtain a matching of $L_j^{(1)}$ and $L_j^{(2)}$. We then repeat this procedure for all $j \in \mathcal{J}_n$. A depiction of the algorithm is provided in Figure 2, and pseudocode is provided in Algorithm 1.

As we shall show, for some $\gamma > 0$, with probability at least $1 - n^{-\gamma}$, we have $T^{(n)}(v, v) < T^{(n)}(v, u)$ for each $u \in L_j^{(2)}$ with $u \neq v$. As a result, whp our algorithm correctly matches every vertex in the top $|\mathcal{J}_n| = \zeta \log_2 n$ layers for some $\zeta > 0$, or equivalently n^α seeds for some $\alpha > 0$. Finally, although it is not essential for subsequent propagation, our analysis also yields a quantitative estimate of α .

Algorithm 1 Finding seeds

- 1: **Input:** graphs G_1, G_2 of size n with layers $L_k^{(1)}, L_k^{(2)}$ revealed $\forall 0 \leq k \leq k_{\max}$, parameter $c_0 > 0$, parameter $\beta \in (2, 3)$, parameter $s \in (0, 1]$
 - 2: **Output:** a set of pairs $\mathcal{S} \subseteq V(G_1) \times V(G_2)$
 - 3: $\mathcal{S} \leftarrow \emptyset$
 - 4: $\alpha \leftarrow 0.009^{\frac{\beta-2}{\beta-1}} (1 - \sqrt{1-s})^2$
 - 5: $\mathcal{J}_n \leftarrow \{k_{\max} - \lfloor \frac{\alpha}{\beta-1} \log_2 n \rfloor \leq j \leq k_{\max}\}$
 - 6: $\mathcal{I}_n \leftarrow \{0 \leq \ell \leq (\beta - 2)k_{\max} - c_0 \log_2 \log_2 n\}$
 - 7: **for** $j \in \mathcal{J}_n$: **do**
 - 8: **for** $v \in L_j^{(1)}$: **do**
 - 9: $v^* \leftarrow \arg \min_{u \in L_j^{(2)}} T^{(n)}(v, u)$ # compute using $\mathcal{I}_n$
 - 10: $\mathcal{S} \leftarrow \mathcal{S} \cup \{(v, v^*)\}$
 - 11: **end for**
 - 12: **end for**
 - 13: **return** $\mathcal{S}$
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Before turning to the analysis of the random variables $T^{(n)}(v, v)$ and $T^{(n)}(v, u)$, we prove a number of general facts. We begin with normal approximations.

Lemma 4.2. *For $n \geq 1$, define the sequence of random variables $Z_n = \frac{1}{\sqrt{N_n p_n}} \sum_{i=1}^{N_n} Y_{i,n}$, where $Y_{i,n} = B_{i,n}^{(1)} - B_{i,n}^{(2)}$ with $B_{i,n}^{(1)}, B_{i,n}^{(2)} \sim \text{Bern}(p_n)$ iid. Suppose $p_n \rightarrow 0$ and $N_n p_n \rightarrow \infty$ as $n \rightarrow \infty$. Then we have $|Z_n| \xrightarrow{d} |N(0, 2)|$ and $Z_n^4 \xrightarrow{d} |N(0, 2)|^4$. As a result, $\mathbb{E}|Z_n| = (1 + o(1))\frac{2}{\sqrt{\pi}}$ and*

$$\mathbb{E}[(|Z_n| - \mathbb{E}|Z_n|)^4] \leq K \tag{7}$$

for sufficiently large n , where $K = 2.1$.

Proof. We first justify $Z_n \xrightarrow{d} N(0, 2)$. Let $X_{i,n} = \frac{1}{\sqrt{N_n p_n}} Y_{i,n}$. Note that $V[Z_n] = 2(1 - p_n) \rightarrow 2$ as $n \rightarrow \infty$. In order to apply the Lindeberg triangular central limit theorem, we must check

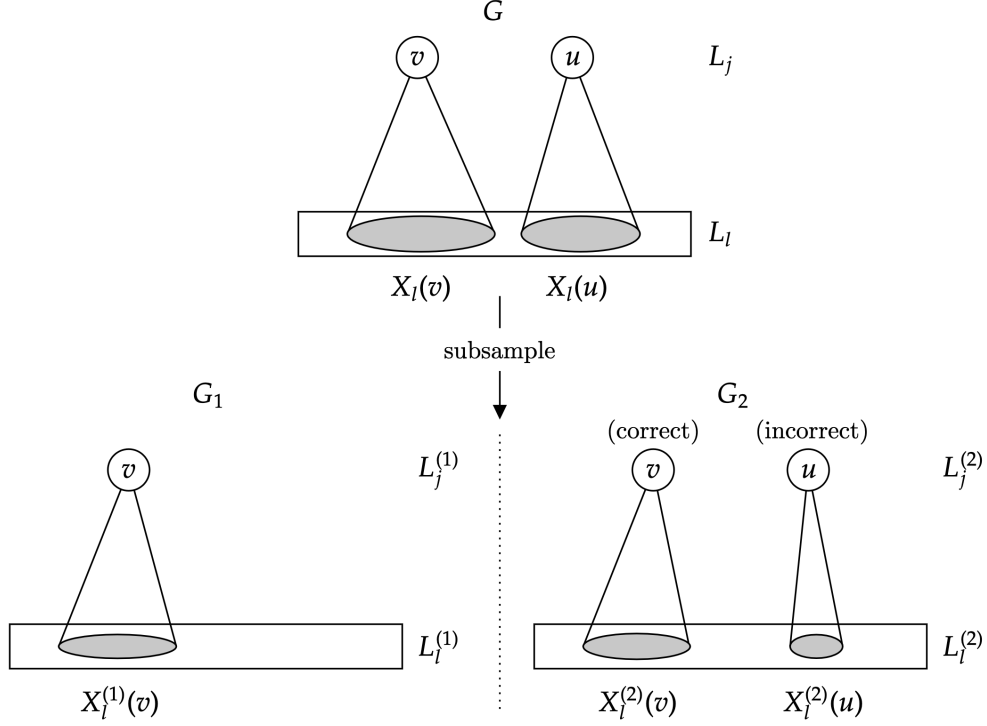


Figure 2: Algorithm for finding seeds. The neighborhoods of a correctly matched pair (v, v) are typically closer in size than the neighborhoods of an incorrectly matched pair (v, u) .

that for all $\varepsilon > 0$,

$$\frac{1}{\sqrt{V[Z_n]}} \sum_{i=1}^{N_n} \mathbb{E}[X_{i,n}^2 \mathbf{1}_{|X_{i,n}| \geq \varepsilon \sqrt{V[Z_n]}}] \rightarrow 0 \quad (8)$$

It suffices to verify the Lyapunov condition $\sum_{i=1}^{N_n} \mathbb{E}[|X_{i,n}|^{2+\nu}] \rightarrow 0$ for some $\nu > 0$ Dembo (2010). We do so for $\nu = 2$. Since $Y_{i,n}$ takes values in $\{-1, 0, 1\}$, we have $\mathbb{E}[X_{i,n}^4] = \mathbb{E}[\frac{1}{\sqrt{N_n p_n}} Y_{i,n}^4] = \frac{1}{N_n p_n} \mathbb{E}[Y_{i,n}^4]$, so that $\sum_{i=1}^{N_n} \mathbb{E}[X_{i,n}^4] = \frac{1}{N_n p_n} V[Z_n] \rightarrow 0$. Thus $Z_n \xrightarrow{d} N(0, 2)$ as $n \rightarrow \infty$.

Next, let $f(z) = z^4$. We claim that the family $(f(Z_n))_{n \geq 1}$ is uniformly integrable; that is,

$$\lim_{M \rightarrow \infty} \sup_{n \geq 1} \mathbb{E}[|f(Z_n)| \mathbf{1}_{|f(Z_n)| \geq M}] = 0 \quad (9)$$

This condition implies that $f(Z_n) \xrightarrow{d} f(N(0, 2))$ as $n \rightarrow \infty$. To verify the condition, it suffices to show that the Lyapunov condition $\sup_{n \geq 1} \mathbb{E}[|f(Z_n)|^{1+\delta}] \leq C$ holds for some $\delta > 0$ and for some constant C . Indeed, if this were the case, then by Markov,

$$\mathbb{E}[|f(Z_n)| \mathbf{1}_{|f(Z_n)| \geq M}] = \mathbb{E}\left[\frac{|f(Z_n)|^{1+\delta}}{|f(Z_n)|^\delta} \mathbf{1}_{|f(Z_n)| \geq M}\right] \leq \frac{1}{M^\delta} \mathbb{E}[|f(Z_n)|^{1+\delta}] \leq \frac{C}{M^\delta} \quad (10)$$

a bound uniform in n that tends to zero as $M \rightarrow \infty$. Since the Lyapunov condition holds for $\delta = \frac{1}{2}$, we deduce $Z_n^4 \xrightarrow{d} N(0, 2)^4$.

We verify the Lyapunov condition for $\delta = \frac{1}{2}$, namely $\sup_{n \geq 1} \mathbb{E}[Z_n^6] \leq C$. Any monomial that appears in the expansion of $(\sum_{i=1}^N Y_{i,n})^6$ is of the form $\prod_{i=1}^6 Y_{i,n}^{\alpha_i}$, where α is a 6-tuple of nonnegative integers such that $\sum_{i=1}^6 \alpha_i = 6$. Note that if α is a 6-tuple such that $\alpha_i \geq 1$ for exactly k indices i , then the expectation of the monomial corresponding to α is $\mathbb{E}[\prod_{i=1}^6 Y_{i,n}^{\alpha_i}] = \prod_{i=1}^k \mathbb{E}[Y_{i,n}^{\alpha_i}] = (2p_n(1-p_n))^k$. Since the distribution of $Y_{i,n}$ is symmetric for all i , any monomial with α_i odd for some i vanishes in expectation. Thus the only nonzero monomials in the expansion correspond to those 6-tuples α such that no entry is less than 2. These correspond to the partitions 6, 4 + 2, and 2 + 2 + 2. The partition 6 corresponds to N_n monomials, the partition 4 + 2 corresponds to $\binom{6}{2} \cdot \binom{n}{2} \cdot 2 \leq 15N_n^2$ monomials, and the partition 2 + 2 + 2 corresponds to $\binom{6}{2} \binom{4}{2} \cdot \binom{n}{3} \leq 15N_n^3$ monomials.

Thus the partition 6 contributes $N_n \cdot 2p_n(1-p_n)$, 4+2 contributes $15N_n^2 \cdot (2p_n(1-p_n))^2$, and 2+2+2 contributes $15N_n^3 \cdot (2p_n(1-p_n))^3$. Since $N_n p_n \rightarrow \infty$ and $p_n \rightarrow 0$, this is bounded by $C(N_n p_n)^3$ for some constant $C > 0$ not depending on n . Thus $\mathbb{E}[Z_n^6] \leq \frac{1}{\sqrt{N_n p_n}^6} \cdot C(N_n p_n)^3 = C$ uniformly in n , as claimed.

Note that the Lyapunov condition above also implies that $|Z_n|^j \xrightarrow{d} |N(0, 2)|^j$ for all $j \in [4]$. In particular $\mathbb{E}|Z_n|^j = (1 + o(1))\mathbb{E}|N(0, 2)|^j$ for $j \in [4]$. Expanding the expression $\mathbb{E}[(|Z_n| - \mathbb{E}|Z_n|)^4]$ and plugging in the asymptotic approximations, we deduce

$$\mathbb{E}[(|Z_n| - \mathbb{E}|Z_n|)^4] = (1 + o(1)) \cdot \sqrt{2}^4 \cdot \mathbb{E}[(|N(0, 1)| - \mathbb{E}|N(0, 1)|)^4] \leq K \quad (11)$$

for $K = 2.1$, where we used the identity $\mathbb{E}[(|N(0, 1)| - \mathbb{E}|N(0, 1)|)^4] = 3 - \frac{12}{\pi^2} - \frac{4}{\pi} \leq 0.52$.

Finally, since $|Z_n| \xrightarrow{d} |N(0, 2)|$, we have $\mathbb{E}[|Z_n|] = (1 + o(1))\mathbb{E}[|N(0, 2)|] = (1 + o(1))\frac{2}{\sqrt{\pi}}$. $\square$

Lemma 4.3. *Let $s \in (0, 1]$ be a constant. For $n \geq 1$, define the sequence of random variables $Z_n = \frac{1}{\sqrt{N_n p_n s}} \sum_{i=1}^{N_n} Y_{i,n}$, where $Y_{i,n} = (B_{i,n}^{(1)} - B_{i,n}^{(2)})E_{i,n}$ are iid, with $B_{i,n}^{(1)}, B_{i,n}^{(2)}, E_{i,n}$ independent, with $B_{i,n}^{(1)}, B_{i,n}^{(2)}$ iid Bern(s), and with $E_{i,n} \sim \text{Bern}(p_n)$. Suppose $p_n \rightarrow 0$ and $N_n p_n \rightarrow \infty$ as $n \rightarrow \infty$. Then if $s < 1$, we have $|\frac{1}{\sqrt{1-s}} Z_n| \xrightarrow{d} |N(0, 2)|$ and $(\frac{1}{\sqrt{1-s}} Z_n)^4 \xrightarrow{d} |N(0, 2)|^4$. As a result, for all $s \in (0, 1]$, $\mathbb{E}|Z_n| = (1 + o(1))\sqrt{1-s}\frac{2}{\sqrt{\pi}}$ and*

$$\mathbb{E}[(|Z_n| - \mathbb{E}|Z_n|)^4] \leq K \quad (12)$$

for sufficiently large n , where $K = 2.1$.

Proof. Since $B_{i,n}^{(1)} - B_{i,n}^{(2)} = \begin{cases} 1 & \text{wp } s(1-s) \\ 0 & \text{wp } 1-2s(1-s) \\ -1 & \text{wp } s(1-s) \end{cases}$, since $E_{i,n} \sim \text{Bern}(p_n)$, and since these

random variables are independent, their product $Y_{i,n}$ takes the values $Y_{i,n} = \begin{cases} 0 & \text{wp } 1-2q_n \\ 1 & \text{wp } q_n \\ -1 & \text{wp } q_n \end{cases}$

where we set $q_n = p_n s(1-s)$. Note $q_n \rightarrow 0$ as $n \rightarrow \infty$.

If $s = 1$, then $Y_{i,n} = 0$ for all i , so that $Z_n = 0$. In this case $\mathbb{E}|Z_n| = 0$ and $\mathbb{E}[(|Z_n| - \mathbb{E}|Z_n|)^4] = 0$, so the two claims hold.

Thus suppose $s \in (0, 1)$ for what follows. Let $W_n = \frac{1}{\sqrt{1-s}} Z_n$. We first justify $W_n \xrightarrow{d} N(0, 2)$. Let $X_{i,n} = \frac{1}{\sqrt{1-s}} \frac{1}{\sqrt{N_n p_n s}} Y_{i,n}$. Note that $V[W_n] = \frac{1}{1-s} \cdot \frac{1}{N_n p_n s} \cdot N_n \cdot 2q_n = 2$. As in the proof of Lemma 4.2, we verify the Lyapunov condition $\sum_{i=1}^{N_n} \mathbb{E}[X_{i,n}^4] \rightarrow 0$. We have $\mathbb{E}[X_{i,n}^4] = \mathbb{E}[\frac{1}{\sqrt{N_n p_n s(1-s)}} Y_{i,n}^4] = \frac{1}{N_n p_n s(1-s)} \mathbb{E}[Y_{i,n}^4]$, so that $\sum_{i=1}^{N_n} \mathbb{E}[X_{i,n}^4] = \frac{1}{N_n p_n s(1-s)} V[W_n] \rightarrow 0$. Thus $W_n \xrightarrow{d} N(0, 2)$ as $n \rightarrow \infty$.

Next, let $f(z) = z^4$. We show that $f(W_n) \xrightarrow{d} f(N(0, 2))$ as $n \rightarrow \infty$. As in the proof of Lemma 4.2, we verify the Lyapunov condition $\sup_n \mathbb{E}[W_n^6] \leq C$ for some constant C . The bound provided in the proof of Lemma 4.2 goes through and yields $\mathbb{E}[(\sum_{i=1}^{N_n} Y_{i,n})^6] \leq C(N_n q_n)^3$ for some constant $C > 0$ not depending on n . Thus $\mathbb{E}[W_n^6] = \frac{1}{\sqrt{1-s}^6} \mathbb{E}[Z_n^6] \leq \frac{1}{\sqrt{1-s}^6} \frac{1}{\sqrt{N_n p_n s}^6} \cdot C(N_n q_n)^3 = C$ uniformly in n , as claimed.

Following the proof of Lemma 4.2, we deduce that $W_n^4 \xrightarrow{d} N(0, 2)^4$, $|W_n| \xrightarrow{d} |N(0, 2)|$, $\mathbb{E}|W_n| = (1 + o(1)) \frac{2}{\sqrt{\pi}}$, and $\mathbb{E}[(|W_n| - \mathbb{E}|W_n|)^4] \leq K$ for sufficiently large n . Plugging in $Z_n = \sqrt{1-s} W_n$, we obtain the result. $\square$

We now provide mgf bounds.

Lemma 4.4. *For $n \geq 1$, define the sequence of random variables $Z_n = \frac{1}{\sqrt{N_n p_n}} \sum_{i=1}^{N_n} Y_{i,n}$, where $Y_{i,n} = B_{i,n}^{(1)} - B_{i,n}^{(2)}$ with $B_{i,n}^{(1)}, B_{i,n}^{(2)} \sim \text{Bern}(p_n)$ iid. Suppose $p_n \rightarrow 0$ and $N_n p_n \rightarrow \infty$ as $n \rightarrow \infty$. Then given $C > 2$ and $c \in (0, 1)$, for sufficiently large n , there exists $\theta_0 > 0$ such that for all $|\theta| \leq \theta_0$, we have $\mathbb{E}e^{\theta(|Z_n| - \mathbb{E}|Z_n|)} \in [c, C]$. In particular, if $C = 2.27$ and $c = 0.9$, then we may take $\theta_0 = 0.10$.*

Proof. For convenience, at times we suppress the subscript n in this proof.

We may write $Z = X_1 - X_2$, where $X_1 = \frac{1}{\sqrt{Np}} \sum_{i=1}^N B_i^{(1)}$ and $X_2 = \frac{1}{\sqrt{Np}} \sum_{i=1}^N B_i^{(2)}$. Note

$$\mathbb{E}e^{\theta X_i} = \left(\mathbb{E}e^{\frac{\theta}{\sqrt{Np}} \cdot \text{Bern}(p)} \right)^N \quad (13)$$

$$= \left(pe^{\theta/\sqrt{Np}} + (1-p) \right)^N \quad (14)$$

Thus by independence,

$$\mathbb{E}e^{\theta Z} = \mathbb{E}e^{\theta X_1} \cdot \mathbb{E}e^{-\theta X_2} \quad (15)$$

$$= \left(pe^{\theta/\sqrt{Np}} + (1-p) \right)^N \cdot \left(pe^{-\theta/\sqrt{Np}} + (1-p) \right)^N \quad (16)$$

To upper bound $\mathbb{E}e^{\theta|Z|}$, note that $e^{\theta|Z|} \leq e^{\theta Z} + e^{-\theta Z}$. Thus

$$\mathbb{E}e^{\theta|Z|} \leq 2 \left(pe^{\theta/\sqrt{Np}} + (1-p) \right)^N \cdot \left(pe^{-\theta/\sqrt{Np}} + (1-p) \right)^N \quad (17)$$

We now examine the terms appearing on the right-hand side. By Taylor expansion of e^x , $p(e^{\theta/\sqrt{Np}} - 1) = p\left(\frac{\theta}{\sqrt{Np}} + \frac{1}{2} \frac{\theta^2}{Np} + O\left(\frac{\theta^3}{(Np)^{3/2}}\right)\right)$. Plugging into the Taylor expansion of $\log(1+x)$,

$$\log\left(1 + p\left(e^{\theta/\sqrt{Np}} - 1\right)\right) = p\left(\frac{\theta}{\sqrt{Np}} + \frac{1}{2} \frac{\theta^2}{Np} + O\left(\frac{\theta^3}{(Np)^{3/2}}\right)\right) + O\left(\frac{\theta^2 p^2}{Np}\right) \quad (18)$$

$$= \frac{\theta p}{\sqrt{Np}} + \frac{1}{2} \frac{\theta^2}{N} + O\left(\frac{\theta^3 p}{(Np)^{3/2}}\right) + O\left(\frac{\theta^2 p^2}{Np}\right) \quad (19)$$

Taking the log of the right-hand side of our bound, the linear terms cancel and we obtain

$$\log \mathbb{E} e^{\theta|Z|} \leq \log 2 + N \log \left(1 + p \left(e^{\theta/\sqrt{Np}} - 1 \right) \right) + N \log \left(1 + p \left(e^{-\theta/\sqrt{Np}} - 1 \right) \right) \quad (20)$$

$$= \log 2 + \theta^2 + O\left(\frac{\theta^3}{\sqrt{Np}}\right) + O(\theta^2 p) \quad (21)$$

$$= \log 2 + \theta^2 + o(1) \quad (22)$$

Next by Lemma 4.2 we have $\mathbb{E}|Z| = (1 + o(1))\frac{2}{\sqrt{\pi}}$, so the log mgf is bounded above by

$$\log \mathbb{E} e^{\theta(|Z| - \mathbb{E}|Z|)} \leq (\log 2 + \theta^2 + o(1)) - (1 + o(1))\frac{2}{\sqrt{\pi}}\theta \quad (23)$$

We now lower bound the mgf. Note that $e^{\theta|Z|} \geq \frac{1}{2}(e^{\theta Z} + e^{-\theta Z})$. Repeating the calculations above, we obtain the lower bound

$$\log \mathbb{E} e^{\theta(|Z| - \mathbb{E}|Z|)} \geq (\theta^2 + o(1)) - (1 + o(1))\frac{2}{\sqrt{\pi}}\theta \quad (24)$$

Fix $C > 2$ and $c < 1$. By continuity of the upper and lower bounds, there exists $\theta_0 > 0$ such that for all $|\theta| \leq \theta_0$, for sufficiently large n , $\log \mathbb{E} e^{\theta(|Z| - \mathbb{E}|Z|)} \in [\log c, \log C]$.

In particular, consider $C = 2.27$ and $c = 0.9$. The inequality $\log 2 + \theta^2 - \frac{2}{\sqrt{\pi}}\theta \leq \log C$ holds for all $|\theta| \leq 0.102$. The inequality $\theta^2 - \frac{2}{\sqrt{\pi}}\theta \geq \log c$ holds for all $|\theta| \leq 0.102$. Thus we may take $\theta_0 = 0.10$ as claimed. $\square$

Lemma 4.5. *Let $s \in (0, 1]$ be a constant. For $n \geq 1$, define the sequence of random variables $Z_n = \frac{1}{\sqrt{N_n p_n s}} \sum_{i=1}^{N_n} Y_{i,n}$, where $Y_{i,n} = (B_{i,n}^{(1)} - B_{i,n}^{(2)})E_{i,n}$ are iid, with $B_{i,n}^{(1)}, B_{i,n}^{(2)}, E_{i,n}$ independent, with $B_{i,n}^{(1)}, B_{i,n}^{(2)}$ iid Bern(s), and with $E_{i,n} \sim \text{Bern}(p_n)$. Suppose $p_n \rightarrow 0$ and $N_n p_n \rightarrow \infty$ as $n \rightarrow \infty$. Then given $C > 2$ and $c \in (0, 1)$, for sufficiently large n , there exists $\theta_0 > 0$ such that for all $|\theta| \leq \theta_0$, we have $\mathbb{E} e^{\theta(|Z| - \mathbb{E}|Z|)} \in [c, C]$. In particular, if $C = 2.27$ and $c = 0.9$, then we may take $\theta_0 = 0.10$.*

Proof. For convenience, at times we suppress the subscript n in this proof.

$$\text{Since } B_i^{(1)} - B_i^{(2)} = \begin{cases} 1 & \text{wp } s(1-s) \\ 0 & \text{wp } 1-2s(1-s) \\ -1 & \text{wp } s(1-s) \end{cases}, \text{ since } E_i \sim \text{Bern}(p), \text{ and since these random}$$

variables are independent, their product Y_i takes the values $Y_i = \begin{cases} 0 & \text{wp } 1-2q \\ 1 & \text{wp } q \\ -1 & \text{wp } q \end{cases}$ where we

set $q = ps(1-s)$. Thus the mgf of Y_i is $\mathbb{E} e^{\theta Y_i} = 1 + q(e^\theta + e^{-\theta} - 2) = 1 + q\theta^2 + O(q\theta^3)$. Since the Y_i are independent, the mgf of the sum is

$$\log \mathbb{E} e^{\theta \sum_i Y_i} = N \log \mathbb{E} e^{\theta Y_i} \quad (25)$$

$$= N \log(1 + (q\theta^2 + O(q\theta^3))) \quad (26)$$

$$= N(q\theta^2 + O(q\theta^3) + O(q^2\theta^4)) \quad (27)$$

by Taylor expansion. Consolidating error terms, replacing θ with $\frac{1}{\sqrt{Nps}}\theta$, and using $Nps \rightarrow \infty$, we obtain

$$\log \mathbb{E}e^{\theta Z} = N \left(q \left(\frac{\theta}{\sqrt{Nps}} \right)^2 + O \left(q \left(\frac{\theta}{\sqrt{Nps}} \right)^3 \right) \right) \quad (28)$$

$$= (1-s)\theta^2 + O \left((1-s) \frac{\theta^2}{\sqrt{Nps}} \right) \quad (29)$$

$$= (1-s)\theta^2 + o(1) \quad (30)$$

With this expansion, we can upper and lower bound $\log \mathbb{E}e^{\theta|Z|}$ exactly as in Lemma 4.4. Specifically, we use the bounds $\frac{1}{2}(e^{\theta Z} + e^{-\theta Z}) \leq e^{\theta|Z|} \leq e^{\theta Z} + e^{-\theta Z}$ to deduce

$$\log \mathbb{E}e^{\theta|Z|} \leq \log 2 + (1-s)\theta^2 + o(1) \quad (31)$$

and

$$\log \mathbb{E}e^{\theta|Z|} \geq (1-s)\theta^2 + o(1) \quad (32)$$

Next by Lemma 4.3 we have $\mathbb{E}|Z| = (1+o(1))\sqrt{1-s}\frac{2}{\sqrt{\pi}}$, so the log centered mgf obeys the bounds

$$\log \mathbb{E}e^{\theta(|Z|-\mathbb{E}|Z|)} \leq \log 2 + ((1-s)\theta^2 + o(1)) - (1+o(1))\frac{2}{\sqrt{\pi}}\sqrt{1-s}\theta \quad (33)$$

and

$$\log \mathbb{E}e^{\theta(|Z|-\mathbb{E}|Z|)} \geq ((1-s)\theta^2 + o(1)) - (1+o(1))\frac{2}{\sqrt{\pi}}\sqrt{1-s}\theta \quad (34)$$

Observe that the upper and lower bounds are the same as in Lemma 4.4, when θ is replaced by $\sqrt{1-s}\theta$. Given $C > 2$ and $c \in (0, 1)$, if $|\theta| \leq \theta_0$, then since $s \in (0, 1]$, we also have $|\sqrt{1-s}\theta| \leq \theta_0$, so that the upper bound is at most $\log C$ and the lower bound is at least $\log c$. The result follows. $\square$

Finally, we provide a subexponential tail bound.

Lemma 4.6. *For each $n \geq 1$, suppose that $Z_1^{(n)}, \dots, Z_{m_n}^{(n)}$ are independent random variables satisfying the following conditions: there exist constants $c, C > 0$ such that*

$$\mathbb{E}e^{\theta(|Z_i^{(n)}|-\mathbb{E}|Z_i^{(n)}|)} \in [c, C] \quad (35)$$

for all $i \in [m_n]$ for all $|\theta| \leq \theta_0$ for sufficiently large n , and there exists a constant $K > 0$ such that

$$\max_{i \in [m_n]} \mathbb{E}[(|Z_i^{(n)}| - \mathbb{E}|Z_i^{(n)}|)^4] \leq K \quad (36)$$

for sufficiently large n . Let $\theta_1 = \frac{x}{2B} \wedge \frac{\theta_0}{2}$, where $B = \frac{(KC^3)^{1/2}}{c^2}$. Then we have the two-sided tail bound

$$\mathbb{P} \left(\left| \frac{1}{m_n} \sum_{i=1}^{m_n} (|Z_i^{(n)}| - \mathbb{E}|Z_i^{(n)}|) \right| \geq x \right) \leq 2 \exp \left(-\frac{1}{2} m_n \theta_1 x \right) \quad (37)$$

Proof. For convenience, at times we suppress the subscript/superscript n in this proof.

We prove upper and lower tail bounds on the average $\frac{1}{m} \sum_{i=1}^m (|Z_i| - \mathbb{E}|Z_i|)$. For convenience, let $W_i = |Z_i| - \mathbb{E}|Z_i|$.

If $x > 0$ and $\theta > 0$, then using the exponential trick and Markov's inequality, we obtain the upper tail bound

$$\mathbb{P}\left(\frac{1}{m} \sum_{i=1}^m W_i \geq x\right) = \mathbb{P}\left(e^{\sum_{i=1}^m \theta W_i} \geq e^{\theta m x}\right) \leq e^{-\theta m x} \prod_{i=1}^m \mathbb{E} e^{\theta W_i} \quad (38)$$

Thus

$$\frac{1}{m} \log \mathbb{P}\left(\frac{1}{m} \sum_{i=1}^m W_i \geq x\right) \leq \frac{1}{m} \sum_{i=1}^m \log \mathbb{E} e^{\theta W_i} - \theta x = f(\theta) \quad (39)$$

We begin by bounding $f''(\theta)$ in a neighborhood of 0. By Cauchy–Schwarz, for $|\theta| \leq \frac{\theta_0}{2}$, our assumptions imply

$$\mathbb{E}[W_i^2 e^{\theta W_i}] \leq \mathbb{E}[W_i^4]^{1/2} \mathbb{E}[e^{2\theta W_i}]^{1/2} \leq (MC)^{1/2} \quad (40)$$

By the dominated convergence theorem, we may interchange differentiation and expectation in $f''(\theta)$ to obtain

$$f''(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{\mathbb{E}[W_i^2 e^{\theta W_i}] \mathbb{E} e^{\theta W_i} - \mathbb{E}[W_i e^{\theta W_i}]^2}{\mathbb{E}[e^{\theta W_i}]^2} \quad (41)$$

for $|\theta| < \frac{\theta_0}{2}$. For the numerator, for $|\theta| \leq \frac{\theta_0}{2}$ this is bounded by $\mathbb{E}[W_i^2 e^{\theta W_i}] \mathbb{E} e^{\theta W_i} \leq (KC)^{1/2} C = (KC^3)^{1/2}$. For the denominator, by assumption, for $|\theta| \leq \frac{\theta_0}{2}$ this is at least c^2 . Since these bounds are uniform in i , $f''(\theta) \leq B$ for $|\theta| < \frac{\theta_0}{2}$, where $B = \frac{(KC^3)^{1/2}}{c^2}$.

We now bound $f'(\theta)$ in a neighborhood of 0. By Cauchy–Schwarz, for $|\theta| \leq \frac{\theta_0}{2}$,

$$\mathbb{E}[W_i e^{\theta W_i}] \leq \mathbb{E}[W_i^2]^{1/2} \mathbb{E}[e^{2\theta W_i}]^{1/2} \leq K^{1/4} C^{1/2} \quad (42)$$

By the dominated convergence theorem we may interchange differentiation and expectation to obtain

$$f'(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{\mathbb{E}[W_i e^{\theta W_i}]}{\mathbb{E} e^{\theta W_i}} - x = -x \quad (43)$$

for $|\theta| < \frac{\theta_0}{2}$. In particular, $f'(0) = -x$. Integrating the inequality $f''(\theta) \leq B$ for $\theta \in [0, \frac{\theta_0}{2}]$, we deduce $f'(\theta) \leq -\frac{x}{2}$ for $\theta \in [0, \theta_1]$, where $\theta_1 = \frac{x}{2B} \wedge \frac{\theta_0}{2}$. Integrating this bound, we deduce $f(\theta) \leq -\frac{1}{2}x\theta$ for all $\theta \in [0, \theta_1]$. We obtain the upper tail bound

$$\mathbb{P}\left(\frac{1}{m} \sum_{i=1}^m (|Z_i| - \mathbb{E}|Z_i|) \geq x\right) \leq \exp\left(-\frac{1}{2}m\theta_1 x\right) \quad (44)$$

Similarly if $x > 0$ and $\theta > 0$, we have the lower tail bound

$$\mathbb{P}\left(\frac{1}{m} \sum_{i=1}^m W_i \leq -x\right) = \mathbb{P}\left(e^{\sum_{i=1}^m -\theta W_i} \geq e^{\theta m x}\right) \leq e^{-\theta m x} \prod_{i=1}^m \mathbb{E} e^{-\theta W_i} \quad (45)$$

Thus

$$\frac{1}{m} \log \mathbb{P} \left(\frac{1}{m} \sum_{i=1}^m W_i \leq -x \right) \leq \frac{1}{m} \sum_{i=1}^m \log \mathbb{E} e^{-\theta W_i} - \theta x = g(\theta) \quad (46)$$

As above, for sufficiently large n , $\mathbb{E} e^{-\theta W_i} \in [c, C]$ for all $i \in m$ for all $|\theta| \leq \theta_0$, and all the above arguments go through. In particular, $g(\theta) \leq -\frac{1}{2}\theta x$ for all $\theta \in [0, \theta_1]$. We obtain the lower tail bound

$$\mathbb{P} \left(\frac{1}{m} \sum_{i=1}^m (|Z_i| - \mathbb{E}|Z_i|) \leq -x \right) \leq \exp \left(-\frac{1}{2} m \theta_1 x \right) \quad (47)$$

Thus for sufficiently large n , for all $x > 0$, we have the two-sided tail bound

$$\mathbb{P} \left(\left| \frac{1}{m} \sum_{i=1}^m (|Z_i| - \mathbb{E}|Z_i|) \right| \geq x \right) \leq 2 \exp \left(-\frac{1}{2} m \theta_1 x \right) \quad (48)$$

□

Having established these general facts, we now specify the subset $\mathcal{I}_n$ of layers that we use to construct our statistic. Fix $c_0 > 0$. We define our set $\mathcal{I}_n$ as

$$\mathcal{I}_n = \{0 \leq \ell \leq (\beta - 2)k_{\max} - c_0 \log_2 \log_2 n\} \quad (49)$$

Each layer $\ell \in \mathcal{I}_n$ obeys condition $(\dagger)$, so that $p_{\ell j}$ is exactly $\frac{2^\ell 2^j}{w_{\text{tot}}}$ for every j . By our definition of $k_{\max}$, the size of $\mathcal{I}_n$ is $|\mathcal{I}_n| = (1 - o(1)) \frac{\beta-2}{\beta-1} \log_2 n$, so that $|\mathcal{I}_n| \geq c_1 \log n$ for any $c_1 \in (0, \frac{\beta-2}{\beta-1})$ for sufficiently large n .

We restrict ourselves to matching seeds in layers j satisfying $j \geq (\beta - 2)k_{\max}$. We shall specify the set $\mathcal{J}_n$ at the very end.

We claim that for each $\ell \in \mathcal{I}_n$ and for each $j \geq (\beta - 2)k_{\max}$, we have $N_\ell p_{\ell j} \rightarrow \infty$ and $p_{\ell j} \rightarrow 0$ as $n \rightarrow \infty$. Since $N_\ell p_\ell \propto 2^{j-(\beta-2)\ell}$ and $p_\ell \propto \frac{2^\ell 2^j}{n} = 2^{\ell+j-(\beta-1)k_{\max}}$, these conditions translate to $j - (\beta - 2)\ell \rightarrow \infty$ and $(\beta - 1)k_{\max} - \ell - j \rightarrow \infty$ respectively. The first condition holds because $j - (\beta - 2)\ell \geq (\beta - 2)c_0 \log_2 \log_2 n$. The second condition holds because $(\beta - 1)k_{\max} - \ell - j \geq c_0 \log_2 \log_2 n$. In fact, we have the uniform control $\min_{\ell \in \mathcal{I}_n} N_\ell p_{\ell j} \rightarrow \infty$.

Fix a pair of vertices $(v, u) \in L_j^{(1)} \times L_j^{(2)}$ with $v \neq u$. We use the above results to show that $T^{(n)}(v, v) < T^{(n)}(v, u)$ whp. We begin by proving tail bounds on each of $T^{(n)}(v, u)$ and $T^{(n)}(v, v)$.

Incorrect match: First, consider the random variables $(|Z_\ell^{(n)}(v, u)|)_{\ell \in \mathcal{I}_n}$.

Recall that $Z_\ell^{(n)}(v, u) = \frac{1}{\sqrt{N_\ell p_{\ell j}}} (X_\ell^{(1)}(v) - X_\ell^{(2)}(u))$, where $X_\ell^{(1)}(v), X_\ell^{(2)}(u)$ are iid $\text{Binom}(N_\ell, p_{\ell j})$.

Let us apply Lemma 4.2 to the sequence of random variables $(\frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, u))_{n \geq 1}$. We deduce that

$$\mathbb{E} \left[\left(\left| \frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, u) \right| - \mathbb{E} \left| \frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, u) \right| \right)^4 \right] \leq K = 2.1 \quad (50)$$

for sufficiently large n . In fact, we can say more. Since $\min_{\ell \in \mathcal{I}_n} N_\ell p_{\ell j} s \rightarrow \infty$, we have the uniform control

$$\min_{\ell \in \mathcal{I}_n} \mathbb{E} \left[\left(\left| \frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, u) \right| - \mathbb{E} \left| \frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, u) \right| \right)^4 \right] \leq K = 2.1 \quad (51)$$

for sufficiently large n .

Fix constants $C > 2$ and $c \in (0, 1)$. If we apply Lemma 4.4 to $\left(\frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, u) \right)_{n \geq 1}$, we deduce that there exists $\theta_0 > 0$ such that for all $|\theta| \leq \theta_0$, we have

$$\mathbb{E} e^{\theta \left(\left| \frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, u) \right| - \mathbb{E} \left| \frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, u) \right| \right)} \in [c, C] \quad (52)$$

for all $\ell \in \mathcal{I}_n$ for sufficiently large n .

Therefore, for each n , we have a collection $\left(\frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, u) \right)_{\ell \in \mathcal{I}_n}$ of random variables that obey the conditions of our tail bound Lemma 4.6. Applying the tail bound, we deduce

$$\mathbb{P} \left(\frac{1}{\sqrt{s}} |T^{(n)}(v, u) - \mathbb{E} T^{(n)}(v, u)| \geq x \right) \leq 2 \exp \left(-\frac{1}{2} |\mathcal{I}_n| \theta_1 x \right) \quad (53)$$

for all $x > 0$ for sufficiently large n .

Correct match: Next, consider the random variables $(|Z_\ell^{(n)}(v, v)|)_{\ell \in \mathcal{I}_n}$.

Recall that $Z_\ell^{(n)}(v, v) = \frac{1}{\sqrt{N_\ell p_{\ell j}}} (X_\ell^{(1)}(v) - X_\ell^{(2)}(v))$, where $X_\ell^{(1)}(v), X_\ell^{(2)}(v)$ are iid $\text{Binom}(X_\ell(v), s)$ conditional on $X_\ell(v)$, and where $X_\ell(v) \sim \text{Binom}(N_\ell, p_{\ell j})$. By definition of the subsampling procedure,

$$X_\ell^{(1)}(v) = \sum_{w \in L_\ell} B_w^{(1)} \mathbf{1}_{(v, w) \in E(G)} \quad (54)$$

and

$$X_\ell^{(2)}(v) = \sum_{w \in L_\ell} B_w^{(2)} \mathbf{1}_{(v, w) \in E(G)} \quad (55)$$

where $B_w^{(1)}, B_w^{(2)}$ are iid $\text{Bern}(s)$, and $\mathbf{1}_{(v, w) \in E(G)} \sim \text{Bern}(p_{\ell j})$ is the indicator that the edge (v, w) is present in the parent graph G .

Subtracting these expressions, we deduce $Z_\ell^{(n)}(v, v) = \frac{1}{\sqrt{N_\ell p_{\ell j}}} \sum_{w \in L_\ell} Y_w$, where we define $Y_w = (B_w^{(1)} - B_w^{(2)}) \cdot \mathbf{1}_{(v, w) \in E(G)}$.

Let us apply Lemma 4.2 to the sequence of random variables $\left(\frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, v) \right)_{n \geq 1}$. We deduce that

$$\mathbb{E} \left[\left(\left| \frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, v) \right| - \mathbb{E} \left| \frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, v) \right| \right)^4 \right] \leq K = 2.1 \quad (56)$$

for sufficiently large n . In fact, we can say more. Since $\min_{\ell \in \mathcal{I}_n} N_\ell p_{\ell j} s \rightarrow \infty$, we have the uniform control

$$\min_{\ell \in \mathcal{I}_n} \mathbb{E} \left[\left(\left| \frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, v) \right| - \mathbb{E} \left| \frac{1}{\sqrt{s}} Z_\ell^{(n)}(v, v) \right| \right)^4 \right] \leq K = 2.1 \quad (57)$$

for sufficiently large n .

Fix constants $C > 2$ and $c \in (0, 1)$. If we apply Lemma 4.4 to $\left(\frac{1}{\sqrt{s}}Z_\ell^{(n)}(v, v)\right)_{n \geq 1}$, we deduce that there exists $\theta_0 > 0$ such that for all $|\theta| \leq \theta_0$, we have

$$\mathbb{E}_e^{\theta} \left(\left| \frac{1}{\sqrt{s}}Z_\ell^{(n)}(v, v) \right| - \mathbb{E} \left| \frac{1}{\sqrt{s}}Z_\ell^{(n)}(v, v) \right| \right) \in [c, C] \quad (58)$$

for all $\ell \in \mathcal{I}_n$ for sufficiently large n .

Therefore, for each n , we have a collection $\left(\frac{1}{\sqrt{s}}Z_\ell^{(n)}(v, v)\right)_{\ell \in \mathcal{I}_n}$ of random variables that obey the conditions of our tail bound Lemma 4.6. Applying the tail bound, we deduce

$$\mathbb{P} \left(\frac{1}{\sqrt{s}} |T^{(n)}(v, v) - \mathbb{E}T^{(n)}(v, v)| \geq x \right) \leq 2 \exp \left(-\frac{1}{2} |\mathcal{I}_n| \theta_1 x \right) \quad (59)$$

for all $x > 0$ for sufficiently large n .

Having proven the tail bounds, we now compare the means $\mathbb{E}T^{(n)}(v, v)$ and $\mathbb{E}T^{(n)}(v, u)$.

By Lemma 4.2, $\mathbb{E} \left| \frac{1}{\sqrt{s}}Z_\ell^{(n)}(v, u) \right| = (1 + o(1)) \frac{2}{\sqrt{\pi}}$, so that $\frac{1}{\sqrt{s}} \mathbb{E}T^{(n)}(v, u) = (1 + o(1)) \frac{2}{\sqrt{\pi}}$.

By Lemma 4.3, $\mathbb{E} \left| \frac{1}{\sqrt{s}}Z_\ell^{(n)}(v, v) \right| = (1 + o(1)) \sqrt{1-s} \frac{2}{\sqrt{\pi}}$, so that $\frac{1}{\sqrt{s}} \mathbb{E}T^{(n)}(v, v) = (1 + o(1)) \sqrt{1-s} \frac{2}{\sqrt{\pi}}$.

Fix $\delta \in \left(0, \frac{2}{\sqrt{\pi}} (1 - \sqrt{1-s})\right)$. For sufficiently large n , we have $\frac{1}{\sqrt{s}} (\mathbb{E}T^{(n)}(v, u) - \mathbb{E}T^{(n)}(v, v)) > \delta$. Our tail bounds imply that for sufficiently large n , the probability that $T^{(n)}(v, v) \geq T^{(n)}(v, u)$ is at most

$$\mathbb{P}(T^{(n)}(v, v) \geq T^{(n)}(v, u)) \quad (60)$$

$$\leq \mathbb{P} \left(\frac{1}{\sqrt{s}} |T^{(n)}(v, v) - \mathbb{E}T^{(n)}(v, v)| \geq \frac{\delta}{2} \right) + \mathbb{P} \left(\frac{1}{\sqrt{s}} |T^{(n)}(v, u) - \mathbb{E}T^{(n)}(v, u)| \geq \frac{\delta}{2} \right) \quad (61)$$

$$\leq 4 \exp \left(-\frac{1}{2} |\mathcal{I}| \theta_1 \frac{\delta}{2} \right) \quad (62)$$

Since $|\mathcal{I}| \geq c_1 \log_2 n$, this is bounded by $4 \exp \left(-\frac{1}{2} \theta_1 \frac{\delta}{2} c_1 \log_2 n \right) \leq n^{-\gamma}$ for some constant $\gamma > 0$.

We now place a lower bound on γ . We set δ close to $\frac{2}{\sqrt{\pi}} (1 - \sqrt{1-s})$, we set C close to 2, and we optimize θ_1 . For sufficiently large n , we can obtain $\theta_1 \geq 0.046 (1 - \sqrt{1-s})$ by setting $c = 0.9$, $C = 2.27$, $\delta = 1.128 (1 - \sqrt{1-s})$, $K = 2.1$, so that $\frac{\delta/2}{2B} \geq 0.046 (1 - \sqrt{1-s})$ and θ_0 can be taken to be 0.10. For sufficiently large n , $c_1 > 0.99 \frac{\beta-2}{\beta-1}$, so our exponent γ is at least

$$\begin{aligned} \gamma &\geq \frac{1}{2} \cdot 0.046 (1 - \sqrt{1-s}) \cdot \frac{1.128 (1 - \sqrt{1-s})}{2} \cdot 0.99 \frac{\beta-2}{\beta-1} \cdot \frac{1}{\log 2} \\ &> 0.018 \frac{\beta-2}{\beta-1} (1 - \sqrt{1-s})^2 \end{aligned} \quad (63)$$

Given $v \in L_j^{(1)}$ and $u \in L_j^{(2)}$ with $u \neq v$ and $j \geq (\beta-2)k_{\max}$, with probability at least $1 - n^{-\gamma}$, the statistic $T^{(n)}(v, v)$ corresponding to the correct match (v, v) is smaller than the statistic

$T^{(n)}(v, u)$ corresponding to the mismatch (v, u) . Given $\alpha < \frac{\gamma}{2}$, a union bound implies that with probability at least $1 - n^{-(\gamma-2\alpha)}$ we can match the set of the n^α highest degree seeds, which constitute the layers $\mathcal{J}_n = \{k_{\max} - \lfloor \frac{\alpha}{\beta-1} \log_2 n \rfloor \leq j \leq k_{\max}\}$.

Thus for $\alpha = 0.009 \frac{\beta-2}{\beta-1} (1 - \sqrt{1-s})^2 \in (0, 0.0045)$, we obtain n^α seeds whp.

4.2 Propagation to log log height

Suppose we have matched n^α seeds whp for some $\alpha > 0$. This corresponds to at least the top $\zeta \log_2 n$ layers in G , where $\zeta = \frac{\alpha}{\beta-1} - o(1)$.

Let $j_{\min}$ denote the height of the lowest fully-matched layer among the seed vertices. In the first stage of propagation, we match the vertices in layers ℓ for $\ell \in [\ell_{\min}, \ell_{\max}]$, where $\ell_{\min}$ and $\ell_{\max}$ will be specified below. For each layer ℓ in this range, we match each pair of vertices based on connections to layer $j_{\min}$. Here $\ell_{\min} \leq \ell_{\max} < j_{\min} - 1$, so that a number of intermediate layers are skipped.

Once layer $\ell_{\min}$ has been matched, we begin the second stage of propagation: we match all layers $\ell \in [\ell_{\min}, \ell_{\min} - 1]$, where $\ell_{\min} \leq \ell_{\min} - 1$ will be specified below. If layer $\ell_{\min} - k$ has already been matched for some integer $k \geq 0$, then the next layer to be matched is layer $\ell = \ell_{\min} - (k + 1)$. We match each pair of vertices in layer ℓ based on connections to the layer $\ell + 1$ directly above.

We now formulate our core routine. The routine takes as input a previously-matched layer L_j , and given $\ell < j$, it outputs a matching of vertices in the layer L_ℓ . The main idea is as follows. Suppose that we want to match a vertex $v \in L_\ell^{(1)}$. If $u \in L_\ell^{(2)}$ is a vertex distinct from v , then u and v will have relatively few common neighbors in layer L_j . However, v and u will have many common neighbors in layer L_j . (Note that it is possible to speak of common neighbors only because we have already matched layers $L_j^{(1)}$ and $L_j^{(2)}$.)

Formally, given a matched layer j and a pair of vertices $(v, u) \in L_\ell^{(1)} \times L_\ell^{(2)}$, define the statistic

$$A_j^{(n)}(v, u) = |\{(w, w) \in L_j^{(1)} \times L_j^{(2)} : (v, w) \in E(G_1) \text{ and } (u, w) \in E(G_2)\}| \quad (64)$$

The quantity $A_j^{(n)}(v, u)$ counts the number of common neighbors of v and u in layer j . By the definition of our model, a correctly matched pair $(v, v) \in L_\ell^{(1)} \times L_\ell^{(2)}$ obeys $A_j^{(n)}(v, v) \sim \text{Binom}(N_j, p_{\ell j} s^2)$, whereas an incorrectly matched pair $(v, u) \in L_\ell^{(1)} \times L_\ell^{(2)}$ obeys $A_j^{(n)}(v, u) \sim \text{Binom}(N_j, p_{\ell j}^2 s^2)$.

Our routine may be stated as follows. Suppose that we have already matched $L_j^{(1)}$ and $L_j^{(2)}$, and suppose that we want to match a vertex $v \in L_\ell^{(1)}$. For each vertex $u \in L_\ell^{(2)}$, compute the quantity $A_j^{(n)}(v, u)$. Let $v^* \in L_\ell^{(2)}$ denote the maximizer $\arg \max_{u \in L_\ell^{(2)}} A_j^{(n)}(v, u)$ of this statistic. We match v^* to v . We repeat this procedure for all $v \in L_\ell^{(1)}$ to obtain a matching of $L_\ell^{(1)}$ and $L_\ell^{(2)}$. An illustration of the routine is provided in Figure 3.

Our algorithm may be stated as follows. In the first stage of propagation, for each layer $\ell \in [\ell_{\min}, \ell_{\max}]$, we apply our routine to the pair $(L_{j_{\min}}, L_\ell)$ to obtain a matching of $L_\ell^{(1)}$ and $L_\ell^{(2)}$. This yields a matching of $L_\ell^{(1)}$ and $L_\ell^{(2)}$ for each $\ell \in [\ell_{\min}, \ell_{\max}]$. In the second stage of propagation, if layer $\ell_{\min} - k$ has already been matched for some $k \geq 0$, then writing

$\ell = \ell_{\text{mid}} - (k + 1)$, we apply our routine to the pair $(L_{\ell+1}, L_\ell)$ to obtain a matching of $L_\ell^{(1)}$ and $L_\ell^{(2)}$. This iterative procedure yields a matching of $L_\ell^{(1)}$ and $L_\ell^{(2)}$ for each $\ell \in [\ell_{\text{min}}, \ell_{\text{mid}} - 1]$. Pseudocode is provided in Algorithm 2.

Algorithm 2 Propagation to log log height

```

1: Input: graphs  $G_1, G_2$  of size  $n$  with layers  $L_k^{(1)}, L_k^{(2)}$  revealed  $\forall 0 \leq k \leq k_{\text{max}}$ , fully-matched
   layer  $j_{\text{min}}$  with  $j_{\text{min}} = k_{\text{max}} - \zeta \log_2 n + O(1)$  for some  $\zeta > 0$ , parameter  $c_0 > 0$ , parameter
    $c_1 > 1$ , parameter  $\beta \in (2, 3)$ 
2: Output: a set of pairs  $\mathcal{S} \subseteq V(G_1) \times V(G_2)$ 
3:  $\mathcal{S} \leftarrow \emptyset$ 
4:  $\ell_{\text{max}} \leftarrow (\beta - 2)k_{\text{max}} - c_0 \log_2 \log_2 n$ 
5:  $\ell_{\text{mid}} \leftarrow (\beta - 2)k_{\text{max}} - (\beta - 2)\zeta \log_2 n + c_1 \log_2 \log_2 n$ 
6:  $\ell_{\text{min}} \leftarrow \frac{c_1}{3-\beta} \log_2 \log_2 n$ 
7:  $\ell \leftarrow \ell_{\text{max}}$ 
8: # stage 1 of propagation
9: while  $\ell \geq \ell_{\text{mid}}$ : do
10:   for  $v \in L_\ell^{(1)}$ : do
11:      $v^* \leftarrow \arg \max_{u \in L_\ell^{(2)}} A_{j_{\text{min}}}^{(n)}(v, u)$ 
12:      $\mathcal{S} \leftarrow \mathcal{S} \cup \{(v, v^*)\}$ 
13:   end for
14:    $\ell \leftarrow \ell - 1$ 
15: end while
16: # stage 2 of propagation
17: while  $\ell \geq \ell_{\text{min}}$ : do
18:   for  $v \in L_\ell^{(1)}$ : do
19:      $v^* \leftarrow \arg \max_{u \in L_\ell^{(2)}} A_{\ell+1}^{(n)}(v, u)$ 
20:      $\mathcal{S} \leftarrow \mathcal{S} \cup \{(v, v^*)\}$ 
21:   end for
22:    $\ell \leftarrow \ell - 1$ 
23: end while
24: return  $\mathcal{S}$ 

```

We now analyze the routine. We place conditions on the layers j and ℓ to ensure a high probability matching.

Suppose that we have already matched the vertices in layer j whp, and suppose that we would like to match the vertices in layer ℓ , where $\ell < j$. First, we require that layer ℓ obeys the condition $(\dagger)$, so that $p_{\ell j}$ is exactly $\frac{2^\ell 2^j}{w_{\text{tot}}}$ for every j . Second, we require $N_j p_{\ell j} \rightarrow \infty$. Third, we require $p_{\ell j} \rightarrow 0$.

Equivalently, we require condition $(\dagger)$, $N_j p_{\ell j} \propto 2^{\ell - (\beta - 2)j} \rightarrow \infty$, and $p_{\ell j} \propto \frac{2^\ell 2^j}{n} = 2^{\ell + j - (\beta - 1)k_{\text{max}}} \rightarrow 0$. The second condition is implied by $\ell - (\beta - 2)j \geq c_1 \log_2 \log_2 n$ for some $c_1 > 1$. The third condition is implied by $\ell + j - (\beta - 1)k_{\text{max}} \leq -c_0 \log_2 \log_2 n$ for some $c_0 > 0$.

We simplify these conditions in two cases.

Case 1: Suppose $j = j_{\text{min}}$. Condition $(\dagger)$ reads $\ell \leq (\beta - 2)k_{\text{max}} - \omega(1)$. The second condition reads $\ell \geq (\beta - 2)j_{\text{min}} + c_1 \log_2 \log_2 n$ for some $c_1 > 1$. Since $j_{\text{min}} = k_{\text{max}} - \zeta \log_2 n +$

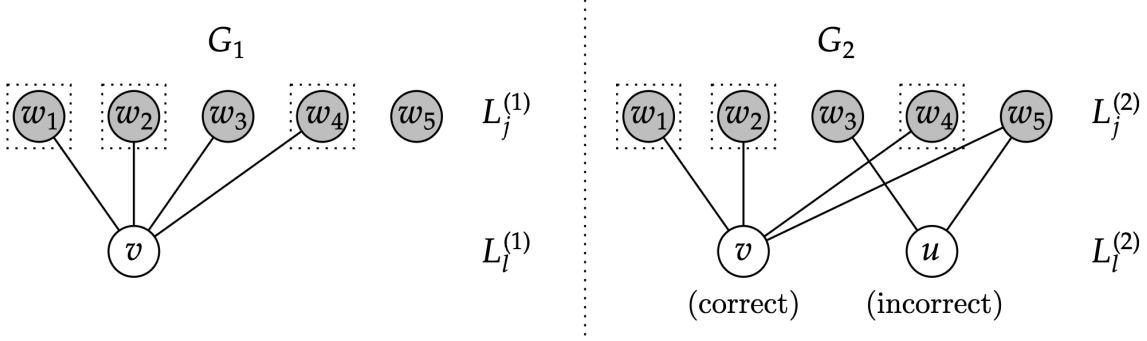


Figure 3: Routine for propagating the matching from the already-matched layer L_j to the layer L_ℓ . The number of agreements between a correctly matched pair (v, v) (here, $A_j(v, v) = 3$) is typically greater than the number of agreements between an incorrectly matched pair (v, u) (here, $A_j(v, u) = 1$).

$O(1)$, this condition can be rewritten as

$$\ell \geq (\beta - 2)k_{\max} - (\beta - 2)\zeta \log_2 n + (1 + o(1))c_1 \log_2 \log_2 n \quad (65)$$

The third condition reads $\ell + j_{\min} \leq (\beta - 1)k_{\max} - c_0 \log_2 \log_2 n$. Plugging in our expression for $j_{\min}$, this condition can be rewritten as

$$\ell \leq (\beta - 2)k_{\max} + \zeta \log_2 n - c_0 \log_2 \log_2 n + O(1) \quad (66)$$

The third condition is rendered obsolete by $(\dagger)$, which is satisfied if the largest ℓ is taken to be

$$\ell_{\max} = (\beta - 2)k_{\max} - c_0 \log_2 \log_2 n \quad (67)$$

for some $c_0 > 0$. Given $c_1 > 1$, up to a factor of $1 + o(1)$, the smallest possible ℓ is

$$\ell_{\text{mid}} = (\beta - 2)k_{\max} - (\beta - 2)\zeta \log_2 n + c_1 \log_2 \log_2 n \quad (68)$$

Note that for sufficiently large n , $\ell_{\text{mid}} \leq (\beta - 2)k_{\max} - \Theta(\log_2 n)$, since $\beta \in (2, 3)$.

Note $\ell_{\max} < j_{\min} - 1$, so that we must skip

$$j_{\min} - \ell_{\max} - 1 = (1 + o(1))((3 - \beta)k_{\max} - \zeta \log_2 n) = (1 + o(1)) \left(\frac{3 - \beta}{\beta - 1} - \zeta \right) \log_2 n \quad (69)$$

many layers in the graph.

Case 2: Now suppose $j = \ell + 1$, where $\ell < \ell_{\text{mid}}$. For $\ell < \ell_{\text{mid}}$, condition $(\dagger)$ is automatically satisfied. The second condition reads $\ell \geq \frac{c_1}{3 - \beta} \log_2 \log_2 n$ for some $c_1 > 1$. The third condition reads $\ell \leq \frac{1}{2} \log_2 n - c_0 \log_2 \log_2 n$ for some $c_0 > 0$. Since $\beta \in (2, 3)$, for $\ell < \ell_{\text{mid}}$ this condition is automatically satisfied for sufficiently large n .

Thus given any constants $c_0 > 0$ and $c_1 > 1$, we shall set $\ell_{\max} = (\beta - 2)k_{\max} - c_0 \log_2 \log_2 n$, $\ell_{\text{mid}} = (\beta - 2)k_{\max} - (\beta - 2)\zeta \log_2 n + c_1 \log_2 \log_2 n$, and $\ell_{\min} = \frac{c_1}{3 - \beta} \log_2 \log_2 n$.

We now estimate the probability that our routine fails for a generic pair of layers L_j and L_ℓ with $\ell < j$.

Lemma 4.7. *Suppose that layer j has already been matched, and let $\ell < j$. Suppose that $N_j p_{\ell j} \rightarrow \infty$ and $p_{\ell j} \rightarrow 0$ as $n \rightarrow \infty$. Consider a vertex $v \in L_\ell^{(1)}$, and consider a vertex $u \in L_\ell^{(2)}$ with $u \neq v$. Then*

$$\mathbb{P}(A_j^{(n)}(v, u) \geq A_j^{(n)}(v, v)) \leq 3 \exp\left(-\frac{1}{3} N_j p_{\ell j} s^2\right) \quad (70)$$

Proof. Recall that $A_j^{(n)}(v, v) \sim \text{Binom}(N_j, p_{\ell j} s^2)$, while $A_j^{(n)}(v, u) \sim \text{Binom}(N_j, p_{\ell j}^2 s^2)$. We use Chernoff to show that $\text{Binom}(N_j, p_{\ell j} s^2)$ is greater than $\text{Binom}(N_j, p_{\ell j}^2 s^2)$ whp.

For the larger binomial, by the two-sided tail bound in Theorem A.1 with $\Delta = (N_j p_{\ell j} s^2)^{\frac{1}{2} + \nu}$ for some $\nu \in (0, \frac{1}{2})$, we deduce $\text{Binom}(N_j, p_{\ell j} s^2) \notin [N_j p_{\ell j} s^2 - \Delta, N_j p_{\ell j} s^2 + \Delta]$ with probability at most $2 \exp\left(-\frac{1}{3} (N_j p_{\ell j} s^2)^{2\nu}\right)$.

For the smaller binomial, we apply the general upper tail bound from Theorem A.1: namely, for all $\varepsilon > 0$ we have $\mathbb{P}(X \geq (1 + \varepsilon)\mu) \leq \exp\left(-\frac{\varepsilon^2}{2 + \varepsilon} \mu\right)$. We apply this inequality to $\text{Binom}(N_j, p_{\ell j}^2 s^2)$ with $\varepsilon\mu = (N_j p_{\ell j} s^2 - \Delta) - N_j p_{\ell j}^2 s^2$. Since $N_j p_{\ell j} \rightarrow \infty$ and $p_{\ell j} \rightarrow 0$, $\varepsilon \rightarrow \infty$, so that $\frac{\varepsilon^2}{2 + \varepsilon} \mu \geq \frac{1}{2} \varepsilon\mu$ for sufficiently large n . We obtain $\text{Binom}(N_j, p_{\ell j}^2 s^2) \notin [0, N_j p_{\ell j} s^2 - \Delta]$ with probability at most $\exp\left(-\frac{1}{2} N_j p_{\ell j} s^2\right)$.

Thus for sufficiently large n , $\text{Binom}(N_j, p_{\ell j}^2 s^2) \geq \text{Binom}(N_j, p_{\ell j} s^2)$ with probability at most $3 \exp\left(-\frac{1}{3} (N_j p_{\ell j} s^2)^{2\nu}\right)$ for any $\nu \in (0, \frac{1}{2})$. Hence given $u \neq v$ in L_ℓ , with probability at most $3 \exp\left(-\frac{1}{3} (N_j p_{\ell j} s^2)^{2\nu}\right)$, $A_j^{(n)}(v, u) \geq A_j^{(n)}(v, v)$. Sending $\nu \rightarrow \frac{1}{2}$ yields the claim. $\square$

Now consider the case that $\ell - (\beta - 2)j \geq c_1 \log_2 \log_2 n$ for some $c_1 > 1$ and $\ell + j - (\beta - 1)k_{\max} \leq -c_0 \log_2 \log_2 n$ for some $c_0 > 0$. We evaluate the error bound given in Lemma 4.7.

We claim that $3 \exp\left(-\frac{1}{3} N_j p_{\ell j} s^2\right) = \frac{1}{n^{2+c_3}}$ for some constant $c_3 > 0$ independent of j, ℓ . To see this, take $\log_2$ of the left-hand side to obtain $\log_2 3 - \log_2 e \cdot \frac{1}{3} N_j p_{\ell j} s^2$. From the second assumption, $N_j p_{\ell j} \geq c_2 2^{c_1 \log_2 \log_2 n} = c_2 (\log_2 n)^{c_1}$ for some $c_2 > 0$. Thus the $\log_2$ of the left-hand side is $O(1) - \log_2 e \cdot \frac{1}{3} \cdot s^2 \cdot c_2 (\log_2 n)^{c_1}$. Since $c_1 > 1$ and since $s > 0$, the last term dominates and the entire expression is at most $-(2 + c_3) \log_2 n$ for some $c_3 > 0$ for sufficiently large n . Thus the left-hand side is at most $\frac{1}{n^{2+c_3}}$, as claimed.

Thus given a matched layer j , an integer $\ell < j$, and a vertex $v \in L_\ell^{(1)}$, the probability that our routine incorrectly matches v is at most $\frac{1}{n^{2+c_3}}$. Since there are at most $N_\ell^2 \leq n^2$ pairs in $L_\ell^{(1)} \times L_\ell^{(2)}$, a union bound implies that the probability that some vertex $v \in L_\ell^{(1)}$ is incorrectly matched is at most $n^2 \cdot \frac{1}{n^{2+c_3}} = \frac{1}{n^{c_3}}$. By one more union bound, the probability that some layer ℓ with $\ell \in [\ell_{\min}, \ell_{\max}]$ is incorrectly matched is at most $(\ell_{\max} - \ell_{\min} + 1) \cdot \frac{1}{n^{c_3}} = O(\log_2 n) \cdot \frac{1}{n^{c_3}} = o(1)$. Thus our algorithm matches all layers in $[\ell_{\min}, \ell_{\max}]$ whp.

4.3 Propagation to constant height

Suppose we have matched all the vertices in layer $\ell_{\min} = \frac{c_1}{3-\beta} \log_2 \log_2 n$ whp for some $c_1 > 1$.

We now perform the third and final stage of propagation: we match all layers $\ell \in [j_{\text{thresh}}, \ell_{\min} - 1]$, where j_{thresh} is of constant order and will be defined below. However, as we enter layers ℓ with $\ell = O(1)$, we can no longer hope to match 100% of the vertices in each

layer whp. Indeed, in the lowest layers of the graph, each vertex has constant degree, so that a positive proportion of the vertices will be isolated. Instead, we merely match a fraction of the vertices in each layer.

We use almost the exact same routine as in the previous stage of propagation. However, we restrict attention to only those vertices that we have correctly matched. Given that we have matched a certain fraction λ_j of the vertices in layer j , we employ the connections between correctly matched vertices in layer j and vertices in layer $\ell = j - 1$ to match a certain fraction λ_ℓ of the vertices in layer ℓ .

Formally, suppose that we have correctly matched the subset $S_j^{(1)} \subseteq L_j^{(1)}$ to the subset $S_j^{(2)} \subseteq L_j^{(2)}$. Given a pair of vertices $(v, u) \in L_\ell^{(1)} \times L_\ell^{(2)}$, define the statistic

$$\tilde{A}_j^{(n)}(v, u) = |\{(w, w) \in S_j^{(1)} \times S_j^{(2)} : (v, w) \in E(G_1) \text{ and } (u, w) \in E(G_2)\}| \quad (71)$$

The quantity $\tilde{A}_j^{(n)}(v, u)$ counts the number of common neighbors of v and u among the correctly matched vertices of layer j . By the definition of our model, if $M_j = |S_j^{(1)}| = |S_j^{(2)}|$, then a correctly matched pair $(v, v) \in L_\ell^{(1)} \times L_\ell^{(2)}$ obeys $\tilde{A}_j^{(n)}(v, v) \sim \text{Binom}(M_j, p_{\ell j} s^2)$, whereas an incorrectly matched pair $(v, u) \in L_\ell^{(1)} \times L_\ell^{(2)}$ obeys $\tilde{A}_j^{(n)}(v, u) \sim \text{Binom}(M_j, p_{\ell j}^2 s^2)$.

Our routine may be stated as follows. Suppose that we have matched the subset $S_j^{(1)} \subseteq L_j^{(1)}$ to the subset $S_j^{(2)} \subseteq L_j^{(2)}$, and suppose that we want to match a vertex $v \in L_\ell^{(1)}$, where $\ell = j - 1$. For each vertex $u \in L_\ell^{(2)}$, compute the statistic $\tilde{A}_j^{(n)}(v, u)$. Let $v^* \in L_\ell^{(2)}$ denote the maximizer $\arg \max_{u \in L_\ell^{(2)}} \tilde{A}_j^{(n)}(v, u)$ of this statistic. If $\tilde{A}_j^{(n)}(v, v^*) \geq 3$, then we match v^* to v ; otherwise, we refrain from matching v . We repeat this procedure for each $v \in L_\ell^{(1)}$ to obtain a partial matching of $L_\ell^{(1)}$ and $L_\ell^{(2)}$. An illustration of the routine is provided in Figure 4.

Our algorithm may be stated as follows. If layer $\ell + 1 = \ell_{\min} - k$ has already been partially matched for some $k \geq 0$, then we apply our routine to the pair $(L_{\ell+1}, L_\ell)$ to obtain a partial matching of $L_\ell^{(1)}$ and $L_\ell^{(2)}$. This iterative procedure yields a partial matching of layer ℓ for each $\ell \in [j_{\text{thresh}}, \ell_{\min} - 1]$. Pseudocode is provided in Algorithm 3.

As we shall see in our analysis below, this criterion is justified because there is a positive probability that the true match $(v, v) \in L_\ell^{(1)} \times L_\ell^{(2)}$ has at least 3 agreements, whereas the probability that some mismatch $(v, u) \in L_\ell^{(1)} \times L_\ell^{(2)}$ at least 3 agreements tends to zero. By means of our procedure, we will be able to match a constant fraction of the vertices in the graph.

Algorithm 3 Propagation to constant height

```

1: Input: graphs  $G_1, G_2$  of size  $n$  with layers  $L_k^{(1)}, L_k^{(2)}$  revealed  $\forall 0 \leq k \leq k_{\max}$ , fully-matched
   layer  $\ell_{\min}$  with  $\ell_{\min} = \frac{c_1}{3-\beta} \log_2 \log_2 n$  for some  $c_1 > 1$ , parameter  $\beta \in (2, 3)$ , parameter  $s \in (0, 1]$ 
2: Output: a set of pairs  $\mathcal{S} \subseteq V(G_1) \times V(G_2)$ 
3:  $\mathcal{S} \leftarrow \emptyset$ 
4:  $C_3 \leftarrow 0.99 \frac{1}{2} s^2 (1 - 2^{-(\beta-2)})$ 
5:  $j_{\text{thresh}} \leftarrow \left\lceil \frac{1}{3-\beta} \log_2 \left( \frac{g^{-1}(0.6)}{0.5 C_3} \right) \right\rceil$  # here  $g(x) = 1 - e^{-x} \left( 1 + x + \frac{x^2}{2} \right)$ 
6:  $\ell \leftarrow \ell_{\min} - 1$ 
7: # stage 3 of propagation
8: while  $\ell \geq j_{\text{thresh}}$ : do
9:   for  $v \in L_\ell^{(1)}$ : do
10:     $v^* \leftarrow \arg \max_{u \in L_\ell^{(2)}} \tilde{A}_{\ell+1}^{(n)}(v, u)$ 
11:    if  $\tilde{A}_{\ell+1}^{(n)}(v, v^*) \geq 3$ : then
12:       $\mathcal{S} \leftarrow \mathcal{S} \cup \{(v, v^*)\}$ 
13:    end if
14:  end for
15:   $\ell \leftarrow \ell - 1$ 
16: end while
17: return  $\mathcal{S}$ 

```

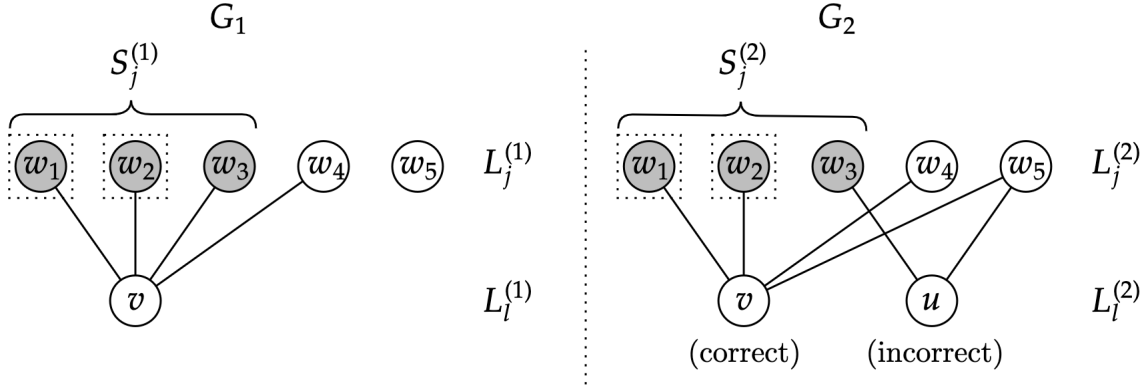


Figure 4: Routine for propagating the matching from the *partially*-matched layer L_j to the layer L_ℓ . Here $S_j^{(i)}$ denotes the correctly matched subset of $L_j^{(i)}$. The number of agreements for a correctly matched pair (v, v) (here, $\tilde{A}_j(v, v) = 2$) is typically greater than the number of agreements for an incorrectly matched pair (v, u) (here, $\tilde{A}_j(v, u) = 1$). However, to ensure a high probability bound, we only match $v \in L_\ell^{(1)}$ to some vertex $v^* \in L_\ell^{(2)}$ if $\tilde{A}(v, v^*) \geq 3$; so that in this case, we refrain from matching v at all.

We now analyze the routine. Let $\lambda_j = \frac{M_j}{N_j}$ denote the fraction of the vertices in layer j that have been matched correctly whp.

Case 1: We first show that the number of agreements $\tilde{A}_j^{(n)}(v, u)$ for a mismatched pair (v, u) is small whp. Specifically, we show that $\mathbb{P}(\text{Binom}(M, q) \geq 2) = O\left(\frac{1}{n^{3-o(1)}}\right)$, where we

use the shorthand $M = \lambda_j N_j$ for the size of M_j , and where $q = p_{\ell_j}^2 s^2$ denotes the probability that v and u share a given neighbor in M_j .

By the formula for the binomial distribution, we have the simple bound

$$\mathbb{P}(\text{Binom}(M, q) = k) = \binom{M}{k} q^k (1 - q)^{M-k} \leq \frac{1}{k!} (Mq)^k \quad (72)$$

Now recall that $Mq = \lambda_j s^2 \cdot N_j p_{\ell_j}^2 = O\left(\frac{2^{(5-\beta)j}}{n}\right)$. Since $j = O(\log_2 \log_2 n)$, we have $Mq = o(1)$, so we may bound

$$\mathbb{P}(\text{Binom}(M, q) \geq 3) \leq (Mq)^3 \left(\frac{1}{3!} + \frac{1}{4!} + \dots \right) = O((Mq)^3) = O\left(\frac{2^{3(5-\beta)j}}{n^3}\right) \quad (73)$$

By a union bound, the probability that $\tilde{A}_j^{(n)}(v, u) \geq 3$ for some mismatched pair $(v, u) \in L_\ell^{(1)} \times L_\ell^{(2)}$ is $O\left(\frac{1}{n^{1-o(1)}}\right)$, as desired.

Case 2: We now show that the number of agreements $\tilde{A}_j^{(n)}(v, v)$ for a true pair (v, v) is at least 3 with positive probability. Specifically, we show that $\mathbb{P}(\text{Binom}(M, q) \leq 2)$ is bounded away from 1, where we use the shorthand $M = \lambda_j N_j$ for M_j , and where $q = p_{\ell_j} s^2$ denotes the probability that v and v share a given neighbor in M_j .

Recall that $Mq = \lambda_j s^2 \cdot N p_{\ell_j} = s^2 C_1 C_2 \frac{1}{2} \lambda_j 2^{(3-\beta)j}$. There exists a constant $C_3 > 0$ such that $Mq \geq C_3 \lambda_j 2^{(3-\beta)j}$ for sufficiently large n . Also note that $q = p_{\ell_j} s^2 = s^2 C_2 \frac{1}{2} \frac{2^{2j}}{n} = \frac{1}{n^{1-o(1)}}$, where the $o(1)$ term is uniform over all $j \leq \ell_{\min}$. Thus $Mq^2 = \frac{1}{n^{1-o(1)}}$, where again the $o(1)$ is uniform over all $j \leq \ell_{\min}$. It follows that $(1 - q)^M = (1 + O(Mq^2))e^{-Mq}$, since $M \cdot \log(1 - q) = -Mq + O(Mq^2)$. We now bound

$$\mathbb{P}(\text{Binom}(M, q) \leq 2) = \mathbb{P}(\text{Binom}(M, q) = 0) + \mathbb{P}(\text{Binom}(M, q) = 1) + \mathbb{P}(\text{Binom}(M, q) = 2) \quad (74)$$

$$= (1 - q)^M + Mq(1 - q)^{M-1} + \binom{M}{2} q^2 (1 - q)^{M-2} \quad (75)$$

$$\leq (1 - q)^{M-2} \left(1 + Mq + \frac{(Mq)^2}{2} \right) \quad (76)$$

$$\leq \left(1 + \frac{1}{n^{1-o(1)}} \right) e^{-Mq} \left(1 + Mq + \frac{(Mq)^2}{2} \right) \quad (77)$$

Thus

$$\mathbb{P}(\text{Binom}(M, q) \geq 3) \geq \left(1 - \frac{1}{n^{1-o(1)}} \right) g(Mq) \quad (78)$$

where $g(x) = 1 - e^{-x} \left(1 + x + \frac{x^2}{2} \right)$. Note that g is increasing on $[0, \infty)$, takes values in $[0, 1)$, and obeys $g(x) < x$ for $x > 0$.

We now show that for sufficiently large n , if $j \leq \ell_{\min}$ with $\lambda_j \geq 0.5$, we have the recurrence $(\star)$

$$\lambda_{j-1} = \left(1 - \frac{1}{n^{1/3}} \right) g(\rho_j \lambda_j) \quad (79)$$

whp, where $\rho_j = C_3 2^{(3-\beta)j}$. That is, so long as we have matched at least 50% of the vertices in layer j , then we can proceed to match a fraction λ_{j-1} of the vertices in layer $j-1$ whp, where λ_{j-1} is given by the expression above.

This fact follows from a simple Chernoff bound. Call a vertex $v \in L_\ell^{(1)}$ good if $\tilde{A}_j^{(n)}(v, v) \geq 3$. Each vertex $v \in L_\ell^{(1)}$ is good with probability at least $r = \left(1 - \frac{1}{n^{1-o(1)}}\right) g(\rho_j \lambda_j)$, and there are $N_\ell = n^{1-o(1)}$ vertices in layer ℓ . By the lower tail bound in Theorem A.1, we have

$$\mathbb{P}(\text{Binom}(N_\ell, r) \leq N_\ell r - (N_\ell r)^{1/2} \cdot 2 \log n) \leq \frac{1}{n} \quad (80)$$

Since we assume $\lambda_j \geq 0.5$, $N_\ell r = O(n^{1-o(1)})$. Thus with probability at least $1 - \frac{1}{n}$, the fraction of good vertices in $L_\ell^{(1)}$ is at least

$$\frac{1}{N_\ell} (N_\ell r - (N_\ell r)^{1/2} \cdot 2 \log n) = \left(1 - \frac{1}{n^{1/2-o(1)}}\right) r \quad (81)$$

$$\geq \left(1 - \frac{1}{n^{1/3}}\right) g(\rho_j \lambda_j) \quad (82)$$

as claimed.

Let j_{thresh} denote the least integer j such that $g(0.5\rho_j) \geq 0.6$. This can be rewritten as

$$j_{\text{thresh}} = \left\lceil \frac{1}{3-\beta} \log_2 \left(\frac{g^{-1}(0.6)}{0.5C_3} \right) \right\rceil \quad (83)$$

Clearly j_{thresh} is of constant order as n tends to ∞ . We propagate our matching down to level j_{thresh} .

We claim that $\lambda_j \geq 0.5$ for all $j \in [j_{\text{thresh}}, \ell_{\min}]$. This is an easy induction on j . The base case is the highest level $j = \ell_{\min}$ defined above. Since all vertices in layer $\ell_{\min}$ are correctly matched whp, we may set $\lambda_{\ell_{\min}} = 1$, which is clearly at least 0.5. Now suppose that $j \in [j_{\text{thresh}} + 1, \ell_{\min}]$ is such that $\lambda_j \geq 0.5$, and let us prove the claim for layer $j-1$. First, note that the recurrence $(\star)$ holds, since $\lambda_j \geq 0.5$. Next, since $j \geq j_{\text{thresh}}$ implies that $\rho_j \geq \rho_{j_{\text{thresh}}}$, and since g is an increasing function, we may bound $\lambda_{j-1} \geq \left(1 - \frac{1}{n^{1/3}}\right) g(\rho_{j_{\text{thresh}}} \cdot 0.5)$. By our definition of j_{thresh} , the right-hand side is at least $\left(1 - \frac{1}{n^{1/3}}\right) \cdot 0.6$, which is at least 0.5 for sufficiently large n . The induction is complete.

We deduce that whp, $\lambda_j \geq 0.5$ for all $j \geq j_{\text{thresh}}$. In particular, our algorithm correctly matches 50% of the vertices in layer j_{thresh} whp. Since $j_{\text{thresh}} = O(1)$, this constitutes $0.50 \cdot C_1 2^{-(\beta-1)j_{\text{thresh}}} \cdot n = \Theta(n)$ vertices. Thus, our algorithm correctly matches a constant fraction of the vertices in the graph whp, and we may conclude.

A Chernoff bound for the binomial distribution

Recall the following version of the Chernoff bound for the binomial distribution; see Theorems 4.4 and 4.5 in Mitzenmacher and Upfal (2017).

Theorem A.1. *If $X \sim \text{Binom}(n, p)$ with mean μ , then $\mathbb{P}(X \geq (1+\varepsilon)\mu) \leq \exp\left(-\frac{\varepsilon^2}{2+\varepsilon}\mu\right)$ for all $\varepsilon > 0$, $\mathbb{P}(X \geq (1+\varepsilon)\mu) \leq \exp\left(-\frac{1}{3}\varepsilon^2\mu\right)$ for all $\varepsilon \in (0, 1]$, and $\mathbb{P}(X \leq (1-\varepsilon)\mu) \leq \exp\left(-\frac{1}{2}\varepsilon^2\mu\right)$ for all $\varepsilon \in (0, 1)$.*

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